Track 2 report

mechanism hypotheses that could raise residual BUBR1 in BUB1B-linked mosaic variegated aneuploidy

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Track 2, Drug Repurposing (SageBio "Rare Disease, Real Kid" MVA Hackathon 2026)

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Abstract

Mosaic variegated aneuploidy (MVA) linked to biallelic *BUB1B* mutations is a chromosome-instability syndrome with growth restriction, microcephaly, severe neurological impairment and childhood cancer predisposition (PMID 15475955, PMID 35804254). Residual BUBR1 protein level sets the severity of checkpoint failure in cell models, and raising the level of a poorly expressed missense BUBR1 mutant to wild-type-comparable levels fully restored the checkpoint response in U2OS cells whose endogenous BUBR1 had been depleted and replaced (PMID 20516114, Fig 4D upper); the companion experiment in cells from people with MVA used wild-type BUBR1 and restored checkpoint activity "almost fully" (PMID 20516114, Fig 1D). This report asks a mechanism question, genotype-agnostic and drug-agnostic: which published mechanisms could raise residual BUBR1 or restore checkpoint function, and which of those are excluded by evidence we can check now? Every result below traces to the pre-registration ledger in Section 12, which carries twenty-one entries with their decision rules hashed before their inputs were opened, with the exceptions disclosed in Section 12.1. One structural test gates everything: a calibrated prediction places the missense allele on the destabilising side of a calibrated cut, with weak evidence, so every abundance-raising mechanism is conditional on a wet assay that has not been done. Eleven of the twenty-five mechanism-class verdict rows are excluded on direction or on dependency measurement, including the one route with a published BubR1-specific rationale: the class I HDAC route was screened across every approved member and excluded on direction on three of the six primary sources examined, the other three cited narrowly with a stated limit each and one of them in fact reporting checkpoint activation, which converts sodium phenylbutyrate's HDAC activity from a second mechanism in its favour into a liability to argue around. PP2A activation is excluded because PP2A-B56 is recruited by BUBR1 and silences the checkpoint. A pre-registered screen of proposed mechanism classes returns twenty-five verdict rows, of which eleven are excluded, six have never been tested against BUBR1 at all and eight are open, six of those eight bearing directly on raising BUBR1; at drug level, 181 molecules were evaluated one by one against the frozen criteria with none kept; it removes the published basis for two of the arms we were carrying and closes two consumer-facing supplements that a reader would otherwise assume were open. Three mechanism families are carried forward, none ordered against the others, and only one of the three is OPEN: for five of seven arms the exposure-versus-engagement arithmetic cannot be done at all, because no source establishes the concentration at which the proposed mechanism acts on BUBR1. We report the negative results in full, the instrument-validity test that declared one of our own criteria non-discriminating, the safety signals in our own cited sources, and a plain-language section for families. Hypotheses for follow-up, not evidence that a medicine works.

This is a living document, edited in place. Nothing in it has been tested in cells from a person with mosaic variegated aneuploidy. Nothing here is a treatment recommendation, a dose, a regimen or medical advice. Where a human exposure appears, it appears only to identify which published human study was admissible as evidence that a mechanism can or cannot be engaged at all.

The Markdown source of this report is in the team's GitHub repository.

AI and data-handling disclosure, user-confirmed and not independently verified. Claude via Claude Code (Anthropic, Claude Max consumer plan, "Help improve Claude" off; these sessions saw child-derived data on local machines only); ChatGPT with Codex, "Sol" (OpenAI consumer app, training off; review and research; model ids measured from the logs, **`gpt-6-luna`** in the box session of 2026-10-10 and **`gpt-6.1-sol`** in the review runs of 2026-10-18); Gemini 3.1 Pro Deep Research (Google business account, data use off; literature research); Perplexity, Grok and DeepSeek V4 (used for general background research and outside critique; no project data was shared with them; plan and data-handling settings were not recorded). No ratings or feedback given; no content flagged for safety review. Model output informed searches and drafts. Every computed number in this report comes from a script whose inputs and decision rules were fixed in the pre-registration log before the script ran, with the exceptions disclosed in Section 12.1.

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1. Summary

This report contains no ranked list of drugs. Every entry below is a **mechanism hypothesis**, stated with the test that kept it and the experiment that would decide it. That is a deliberate change from the previous revision, which published an ordinal list; the ordinal output is withdrawn (user decision, 2026-10-20), and the two tests reported here independently support withdrawing it.

Seven results shape the answer.

1. **There is a gate, and it is not a drug question.** A pre-registered structural test places the missense *BUB1B* allele on the destabilising side of a calibrated cut, evidence **weak** (Section 3). Every abundance-raising mechanism in this report is conditional on that gate, and the wet assay decides it, not the prediction. If the allele is function-defective rather than abundance-limited, raising protein level could be useless or actively harmful, and no falsifier we had written covered a worse-with-more outcome. One is now written.
2. **One mechanism family is excluded on direction by the source that had been cited in its favour.** PP2A activation is out. The activating action belongs to the unphosphorylated species, which is the least available one at labelled exposure; and PP2A-B56 is recruited *by* BUBR1 and, at the kinetochore, extinguishes the checkpoint (PMID 23926298, PMID 33207204, PMID 23079597, PMID 23789096). Driving the silencing step in a cell that already fails to hold the checkpoint points the wrong way.
3. **The most promising mechanism available to us is excluded on direction.** HDAC2 and HDAC3 bind acetylated BubR1 and initiate its degradation (PMID 28985013), which predicts that inhibiting them stabilises BubR1. We pre-registered a screen of every approved class I HDAC inhibitor. **Zero entrants.** The class is excluded on direction on three of the six primary sources examined, in human cells, transcription-independently, and class I selectivity does not escape it (Section 8.2); the other three sources are cited narrowly with a stated limit each, one of them in fact reports checkpoint **activation**, and the class verdict does not depend on any of those three. The consequence for an existing candidate is direct: the HDAC-inhibitory component of sodium phenylbutyrate is a risk to argue around, not a second mechanism in its favour, and its total plasma concentration is in the order where that activity occurs, although by the pre-registered rule the free concentration is UNRESOLVED and no "reaches" verdict is recorded (Section 8.2).
4. **For most surviving arms the engagement arithmetic cannot be done at all.** A symmetric exposure test covered seven arms. Five have no establishable concentration at which the proposed mechanism was demonstrated on BUBR1 (Section 7.1). The surviving mechanisms are therefore **not ordered by engagement evidence**, and we do not order them.
5. **Criterion C4 does not discriminate, so it clears and excludes nothing.** Criterion C4, which compares a plasma concentration with an intracellular enzyme's IC50, is declared **NON-DISCRIMINATING** (Section 9.3). It removes nothing retroactively. It does mean the nicotinamide exclusion is not evidence that nicotinamide is harmful, and the NMN and NR pass is not evidence that they are safe.

6. **One negative result is powered against the effect size it pre-specified; the other's threshold excludes its own positive control.** `T2-DEPMAP-01` is not relabelled: at its own threshold the test had power 1.000 against the effect size it pre-specified, and 5 of 5 positive controls passed. `PRE-REG-B` **is** relabelled: its threshold excludes its own BUB1 positive control, so its negative is a property of the rule, not of the biology (Section 9.4).
7. **A screen of twenty-five mechanism-class verdict rows downgraded two arms carried here and closed two supplements sold in every shop** (Section 15). A fixed search pairing BUBR1 with chemical chaperones returned **zero records**, so the rationale we carried for sodium phenylbutyrate has no source connecting it to BUBR1 at all, and that arm is downgraded to MECHANISM NOT TESTED. A later citation check found that the heat-shock class had been raised to OPEN on PMID 30496702, but that paper's own proposed mechanism runs through Cdc20, it reports BubR1's association with Cdc20 as unaffected by HSPH1 knockdown, and its only BubR1-specific observation is conditioned on heat shock. That arm is moved back down to MECHANISM NOT TESTED, and the reversal is reported rather than the upgrade kept (Section 7.3). Curcumin and vitamin C both close as MECHANISM NOT TESTED, each for reasons set out in full, because a reader who found them in a candidate universe would otherwise assume they were live. Of twenty-five verdict rows, **11 are excluded, 6 are not tested and 8 are open**, and six of the eight open rows bear directly on raising BUBR1 or restoring the checkpoint.

Mechanism hypotheses carried forward (Section 7, listed alphabetically; the order carries no meaning): chemical-chaperone proteostasis, **NOT TESTED** (sodium phenylbutyrate, prescription); heat-shock co-induction, **NOT TESTED** (arimoclomol, prescription); NAD+ raising into a SIRT2-dependent step, **OPEN** (niacin, prescription; NMN and NR, dietary supplements that are not approved medicines). **Mechanism families excluded (Sections 8 and 15):** PP2A activation; class I HDAC inhibition; mTOR inhibition; APC/C inhibition; other mitotic-target inhibition (PLK4, KIF18A, AURKB, PLK1); TTK/MPS1 inhibition; sirtuin inhibition; lysine acetyltransferase inhibition; microtubule-targeting agents; DNA-damaging and topoisomerase cytotoxics; the SIRT2 shared-stability route at fitness level; any pre-specified combination lever.

No mechanism is proposed as effective, and no molecule named here is proposed for use by anyone. None has been tested in MVA cells by us or, as far as our searches reach, by anyone.

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2. What the condition is, and what an intervention could plausibly touch

This section exists because the previous revision had no clinical description, and without one a reader cannot judge what any of the mechanisms below would be for.

Clinical picture. MVA is a rare recessive condition "characterized by growth retardation, microcephaly, childhood cancer and constitutional mosaicism for chromosomal gains and losses" (PMID 15475955, verbatim, verified at abstract). The *BUB1B*-linked form (MVA1) is described clinically as intrauterine growth restriction, post-natal growth retardation and severe neurological impairment including microcephaly, developmental delay or intellectual disability, epileptic seizures and generalised hypotonia, with malignancy as a serious sequela; a 2022 review assembled 31 reported MVA1 cases plus four prenatally diagnosed ones (PMID 35804254, verified at abstract). Premature chromatid separation (PCS) is present in most cases and is the cytogenetic finding a clinician actually sees (PMID 18932004, verified at abstract).

Cancer predisposition. This is the part that matters most for safety assessment. Hanks 2004 identified truncating and missense *BUB1B* mutations in five MVA families, "including two with embryonal rhabdomyosarcoma", and framed the finding as "constitutional aneuploidy and cancer predisposition" (PMID 15475955, verified at abstract). Related MVA loci carry the same signal: biallelic *TRIP13* mutations "predispose to Wilms tumor and chromosome missegregation" (PMID 28553959, verified at title and record level).

What is already fixed, and what is not. Growth restriction, microcephaly and the structural neurological phenotype are developmental and are largely established by the time a diagnosis exists. The component that is still ongoing is the generation of **new** missegregation events in dividing tissue, and therefore the ongoing tumour risk. The honest statement of goal for every mechanism in this report is: reduce the rate of new missegregation. We did not find, and do not propose, any mechanism with evidence that it reverses an established developmental phenotype.

A patient-followable readout does not yet exist here. The closest candidate is the PCS or aneuploidy rate on a peripheral blood or fibroblast karyotype, which is already measured diagnostically. We flag as an open requirement that no mechanism in this report has a validated biomarker in this disease, and that a PCS rate has not been shown to respond to any intervention.

Locus heterogeneity. MVA is genetically heterogeneous: *CEP57* causes MVA2 (PMID 21552266, verified at title and record level) and *TRIP13* causes MVA3 (PMID 28553959). This matters twice. First, nothing in this report generalises beyond the *BUB1B* mechanism. Second, *TRIP13* is used in Section 9 as a statistical positive control, and it is itself an MVA gene; that was not stated before and is stated now.

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3. The gate: is the missense allele abundance-limited?

Every mechanism in Section 7 raises the amount of BUBR1 protein or slows its loss. All of them are conditional on this one question, and it is not a drug question. It comes first for that reason.

Result, in the adversarial verifier's own recommended wording (`PRE-REG-R2A`, verification VERIFY-R2A, quoted verbatim):

> A pre-registered structural test (ThermoMPNN ddG on the AlphaFold model, calibrated on PTEN and TPMT VAMP-seq abundance data, pooled AUC 0.86) places the child's missense allele on the destabilising side of the calibrated cut (1.51 against 1.15 kcal/mol), consistent with the published low-abundance class that was rescued by raising protein level. The evidence is weak and does not establish the premise: at that ddG the calibrated probability of low abundance is 0.53 (95% CI 0.49 to 0.58) against a 0.43 base rate, the 0.36 kcal/mol margin is smaller than the method's published error (RMSE 0.71 to 1.52 kcal/mol), and on BUBR1's own published missense alleles the method shows no demonstrated discrimination (AUC 0.60, 95% CI 0.08 to 1.00). Every abundance-raising candidate therefore remains conditional on the wet assay: steady-state level, cycloheximide chase, and rescue at matched expression, with Q921H as an in-domain wild-type-level control.

What the verification found. The computation is exactly reproducible (all 58,760 ddG values at maximum absolute difference 0.000000 through an independent code path) and the pre-registration is intact (hash-verified, file mode 444, mtime chain consistent). Three things do not survive as support. First, of the five published low-abundance BUBR1 alleles, only three are called destabilising at the cut; had the pre-registration named a different control pair, the outcome would have been UNRESOLVED, so the in-protein gate is "a 30 percent coin". Second, one annotation of the kinase-domain boundary makes a published low-abundance allele "far" from the domain, so the domain-proximity criterion is not discriminative and is not cited as support. Third, the freeze log is self-attested on one machine with no external notarisation; the hash agreement and mtime chain are consistent with the claimed order but cannot prove it to a hostile reader. An independent second opinion (ESM-2 650M, PMID 36927031, pooled AUC 0.840 on the same calibration sets) agrees on direction for this allele and reproduces the same failure pattern on BUBR1's published alleles; two partly redundant methods agreeing is mild corroboration, not evidence.

The instruments behind this gate are named without citations, and that is a gap in this report rather than in the method. ESM-2 is cited above and in the Track 1 methods report. The other three carry no literature citation anywhere in this document and are marked **unsourced** here: the ddG predictor

(ThermoMPNN), the structure it was run on (an AlphaFold model, obtained from AlphaFold DB as Section 12.2 records) and the two abundance calibration sets (PTEN and TPMT VAMP-seq, obtained from MaveDB). The versions and accessions are in the pre-registration log; identifiers for their publications were not resolved for this revision and are not invented here. A methods reviewer should treat the gate's instruments as uncited until that is done.

Pre-registered falsifier of the gate, and the one that was missing. The gate fails if the wet assay shows the missense product at wild-type-comparable steady-state level and wild-type-comparable half-life. The falsifier that did not exist before and now does: **if the allele is function-defective rather than abundance-limited, raising protein level could be useless or harmful.** A destabilised-but-folding or a folded-but-inactive BUBR1 raised in abundance would still occupy Cdc20 and Bub3 while failing to support kinetochore-microtubule attachment, which is a dominant-negative concern, not merely a null result. **No falsifier anywhere in the previous revision covered a worse-with-more outcome.** The decisive experiment is therefore not a half-life measurement alone: it is rescue at matched expression, scored on chromosome alignment and mitotic index, with a dose-response arm that can detect a non-monotonic response.

A folding-stability predictor cannot see the whole question. Acetylation-linked BubR1 turnover is on record at two residues, K250 (PMID 19407811, PMID 25285631) and K668 (PMID 24825348), and a substitution can create or remove a modification site without changing fold stability. A ddG model is blind to a new ubiquitin or acetyl acceptor site. The identity of the substituted residue is held to genotype class and is not reported here. The pre-registration says so, and the verification could not improve on it.

Relationship to the earlier placement test. This result does not reverse `PRE-REG-D`, which asked a different question (pathogenicity placement, where two of three calibrated predictors contradicted the placement). R2-A asks about abundance class with an instrument built for abundance. Both are predictions. Neither is a measurement.

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4. Mechanism and the evidence behind it

4.1 *BUB1B* and BUBR1. Hanks 2004 identified truncating **and missense** *BUB1B* mutations in five MVA families (PMID 15475955, verified at abstract). BUBR1 contributes to kinetochore-microtubule attachment and to the mitotic checkpoint. Cells from MVA patients with biallelic mutations have an impaired mitotic checkpoint, chromosome alignment defects and low overall BUBR1 abundance (PMID 20516114, verified in full text).

An animal model of the genotype class exists, and it belongs in this evidence base. Sieben 2020 mimicked the BUBR1 L1012P substitution in mice, as BubR1 L1002P, and combined it with two further MVA variants, BUBR1 X753 (a truncated protein) and BUBR1 H (low amounts of wild-type protein). BubR1 X753/L1002P mice died prematurely; BubR1 H/L1002P mice were viable and showed many MVA features including cancer predisposition and progeroid pathologies, and several of those pathologies were attenuated in severity relative to BubR1 H/H mice at a similar total BUBR1 reduction. The paper's conclusion is that "BUBR1 allelic effects beyond protein level and aneuploidy contribute to disease heterogeneity" (PMID 31738183, verified at abstract; the paper carries an erratum, *J Clin Invest* 2020;130(11):6188, which we have not read). Two consequences for this report. The evidence base is not cell culture alone: the dose-to-phenotype relation Section 4.2 anchors in a HeLa knockdown series has an in vivo counterpart, and L1012P is one of the two alleles the rescue in Section 4.3 was demonstrated on. And the same conclusion cuts against a pure abundance account, because it attributes part of the phenotype to allele identity rather than to protein level, which is the gate question of Section 3 restated in a mouse. The model is cited as evidence base and as a caution, never as evidence that a mechanism below works.

4.2 Residual BUBR1 is dose-dependent, and PCS appears before aneuploidy. Bohers 2008 produced stable shRNA HeLa lines with residual BUBR1 at 8.5, 10, 14, 58 and 77%. Verbatim: "Induction of PCS was

detected in all transduced cells and its level was correlated to the decrease of BUBR1" (PMID 18932004, verified at abstract). The same abstract reports aneuploidy detected in cells with residual BUBR1 below 50%. Two readings follow, and the previous revision gave only the second. First, **PCS was induced at every level tested, including 58% and 77% residual**, so the cytogenetic phenotype is not abolished at high residual protein and the effective target for a cytogenetic endpoint sits above 50%. Second, the aneuploidy statement is an interval claim: the gradient has no measured point between 14% and 58%. This is a HeLa shRNA design anchor, not a patient threshold and not a dose.

4.3 The strongest published support for an abundance strategy. Forced overexpression of the poorly expressed substitutions I909T and L1012P to levels comparable to wild-type BUBR1 **fully restored the response to nocodazole** in U2OS cells whose endogenous BUBR1 had been depleted by shRNA and replaced by an RNAi-insensitive construct (PMID 20516114, Fig 4D upper, verified in full text; every panel of that figure is U2OS or HeLa, and none uses patient cells). The companion experiment in cells from people with MVA is Fig 1D of the same paper: it used wild-type BUBR1 rather than the mutants, and checkpoint activity was "almost fully restored to parental levels upon re-expression of BUBR1 in the patient cells" (verbatim, verified in full text). The two differ in reach and in strength, and keeping them apart makes this report's standing statement that nothing has been tested in MVA cells by us more consistent, not less. Delivery was ectopic LAP-tagged BUBR1 cDNA by plasmid cotransfection, not mRNA. The same paper verifies the mechanism that makes a stability instrument the right instrument: low abundance of the substitution mutants depends on HSP90, is prevented by MG132, and is "at least in part, caused by decreased protein stability". **The effect size that restored function was wild-type-comparable, not 1.25-fold.** Section 10 states the consequence for our own go/no-go proposal.

4.4 The SIRT2 abundance lever, and where it is contested. North 2014 reports that NMN raised BubR1 protein in wild-type mouse embryonic fibroblasts and that the induction was "almost completely absent" in Sirt2-knockout MEFs (PMID 24825348, verified in full text, Fig 6F and 6G), plus a separate in vivo arm in mouse testes (Fig 6D). That paper tested **NMN only**: it did not test niacin or nicotinamide riboside. Counter-evidence: Suematsu 2014 reports that SIRT2 knockdown increases BubR1 acetylation at prometaphase but that this "is not substantial to elevate the levels of total BubR1" (PMID 25285631, verified at abstract; full text closed access). The two papers assign acetylation **opposite signs at different residues**: North has CBP acetylation at K668 driving BubR1 to degradation, so SIRT2 deacetylation stabilises; Suematsu has acetylation at K250 inhibiting APC/C-dependent proteolysis, so acetylation stabilises. A K250 result cannot replicate or fail to replicate a K668 result. SIRT2 knockdown removes activity at both residues, so the disagreement on the knockdown result itself is not dissolved by the residue argument.

4.5 Correction: the corroborating source is independent, and it supports the axis. The previous revision said the only other source reporting the abundance effect "shares authors with PMID 24825348" and used it to argue single-laboratory dependence. That was wrong in the direction against our own interest, and it is corrected here. **PMID 25672491 is Rumpf T et al, *Nat Commun* 2015, a 19-author paper, first author Rumpf and last author Jung (Freiburg); North and Sinclair are 2 of those 19** (verified at record and abstract level, 2026-10-20). It is not a same-laboratory echo. Its content is pharmacological, not genetic: a selective SIRT2 inhibitor "induces destabilization of the checkpoint protein BubR1" (verbatim). That is independent pharmacological support for the SIRT2-to-BubR1 stability axis in the direction this report needs, obtained with a tool compound rather than a knockout. The accurate shared-author statement is the one above; the claim of single-lab dependence is withdrawn. This does not revive the fitness-level route, which is separately falsified (Section 9.4): a pharmacological destabilisation result and a cross-cell-line fitness correlation are different claims, and only the second was tested by DepMap.

4.6 Correction: the PP2A-B56 claim is no longer unverified. The previous revision marked "PP2A-B56 recruitment matters for BUBR1 function" as its one self-declared unverified claim. It is verified, in three primary sources, and the verification is what excludes the PP2A mechanism (Section 8.1): PP2A-B56 is "recruited to the kinase attachment regulatory domain (KARD) of … BUBR1 in a phospho-dependent

manner", unleashing a relay "extinguishing the checkpoint and promoting anaphase" (PMID 33207204, verbatim); "removal of BUBR1 from mitotic cells … reduces PP2A-B56alpha kinetochore binding" (PMID 23079597, verbatim); and BUBR1 recruits PP2A via B56 to promote chromosome congression, with failure of that recruitment contributing to the congression defects in MVA patient-derived cells (PMID 23789096, verified at abstract). The "unverified" tag is retired.

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5. Scope and constraints

- **Genotype class only.** The class is one truncating and one missense *BUB1B* allele. The trans configuration is **inferred, not measured**: read-backed phasing returned UNRESOLVED because no sequencing template in the data spans the gap between the two sites (P1 RESULT). No child-derived coordinate, genotype string, phenotype detail or score appears in this report.
- **Mechanism-first, genotype-agnostic evaluation.** Every drug or supplement named below was evaluated against the published mechanism: does the literature support or exclude this mechanism as a way to raise BUBR1 level or restore checkpoint function in cell models? Verdicts are about mechanisms and evidence, never about a person.
- **No ordinal ranking.** Mechanism hypotheses are listed alphabetically. Position says nothing about promise, safety or priority.
- **Scope of molecules.** Track 2 asks about market-approved medications. Dietary supplements are named where they carry the published mechanism rationale and are labelled "dietary supplement, not an approved medicine". Research-stage ideas are listed separately.
- **Region-specific approval.** "Approved" is stated per region. US and EU status differ materially for two molecules.
- **Paediatric labelling is reported, not used to exclude.**
- **Blood-lineage dependency is checked for every candidate target DepMap measures.** 24 of 26 pre-registered genes returned values; HSPA1A and HSPA1B have no column in CRISPRGeneEffect.csv and are reported as "not measured", never as zero.
- **Exposure figures are archival.** Where a human dose appears it identifies the published study that produced the concentration. It is not a proposal.

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6. How mechanisms were evaluated

Four instruments, each pre-registered before its inputs were opened.

6.1 Evidence grade (descriptive only). G1: human data on BUBR1 or aneuploidy endpoints in a clinical or patient-derived setting, replicated. G2: patient-derived cell data on BUBR1 protein or mitotic endpoints, one or more independent sources. G3: mouse or cell-line data on BUBR1 protein or checkpoint endpoints, not contradicted by an independent source. G4: mechanism or pathway evidence only, or G3 evidence with a published contradiction. **Every mechanism in Section 7 is G4, with one contested exception.** The grade therefore separates nothing, and it is reported as a label rather than as an ordering device. The ranking rule that formerly consumed these grades, its misapplication and its correction are in Appendix A; its ordinal output is withdrawn.

6.2 On-target dependency (`T2-P4`, `T2-P5`). Fraction of DepMap 24Q4 lines with Chronos gene effect below -0.5, flagged at or above 0.25, with blood-lineage medians reported. Descriptive. A knockout dependency is a loss-of-function proxy in cancer lines; it is not a safety measure for a drug and not a measure of anything in normal marrow.

6.3 Mechanism direction (criterion C4, and its symmetric successor `T2-EXPOSURE-01`). C4 asks whether a molecule's reached human exposure opposes the mechanism it is proposed for. It was applied asymmetrically in the previous revision, to the NAD-route molecules only. `T2-EXPOSURE-01` applies the

question symmetrically to all seven arms (Section 7.1) and is the instrument that excludes the PP2A route. C4 itself has since failed a validity test and is reported as non-discriminating (Section 9.3).

6.4 Mechanism-class screen (`T2-HDAC-R2C`). For a mechanism with a published BubR1-specific rationale, screen every approved member of the drug class against regulatory status, dependency, exposure, direction and evidence grade, with the falsifier written in advance. Run once, on the class I HDAC route. Outcome: zero entrants (Section 8.2).

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7. Mechanism hypotheses carried forward

Three families. None is ordered against the others, and Section 7.1 is the test that establishes there is no basis to order them. Every entry is conditional on the gate in Section 3.

Where this set came from, stated because no rule selects it. These three families are the candidate list this project was already carrying when the ordinal output was withdrawn, less the two entries that were removed from it: fingolimod, excluded on mechanism direction and exposure (Section 8.1), and nicotinamide, dropped by a frozen criterion (Section 8.3). The provenance is in the ledger (`RANKING-RULE-T2` and `APPLICATION-FIX-T2-01`, Section 12). No pre-registered rule selects three families out of the twenty-five verdict rows of Section 15, and this is not the set of open classes: two of the three here are NOT TESTED, while six of Section 15's eight open rows bear directly on raising BUBR1 and are not carried forward in this section. A reader should treat Section 7 as the hypotheses this work happened to carry, with the test that kept each one, and Section 15 as the wider screen, rather than reading Section 7 as a selection out of Section 15.

7.1 Exposure and target engagement, applied symmetrically (`T2-EXPOSURE-01`)

Pre-registered and frozen before any figure below was opened. "`C_demo`" is the concentration at which the proposed mechanism was demonstrated on BUBR1 or BubR1 in the cited source. [V] verified at source; [U] not established. Doses identify the published human study only.

Arm	Published human exposure	Achieved free concentration	`C_demo` on BubR1	Ratio	First step in target cells	Direction and outcome
PP2A activation (fingolimod)	0.5 mg/day oral, label PK [V]	fingolimod-phosphate 3.48 nM total, >99.7% bound, 86% of parent in red cells: free below 10.4 pM [V]	not established: the source does not measure BUBR1 at all [U]; low micromolar in that literature	about 1e-5 at `C_demo` 1 uM; about 1e-4 even at 100 nM	fails earlier	**AGAINST** [V]. **E-FAIL on both limbs** (the frozen exposure rule's failure band, a ratio below 0.01; the bands are set out in Appendix B class 16b). **EXCLUDED** (Section 8.1)
Chemical chaperone (4-PBA)	label single-dose PK, 5 g oral: Cmax 218 ug/mL tablets, 195 ug/mL powder [V]	not computable; label states disposition "has not been characterized fully", so no protein binding figure exists. Total 1.33 mM [V]	not established: the source is an alpha1-antitrypsin paper and does not measure BUBR1 at all [U]	not computable	BUBR1 never shown to be a chemical-chaperone client [U]	UNKNOWN. **E-UNRESOLVED**

Arm	Published human exposure	Achieved free concentration	`C_demo` on BubR1	Ratio	First step in target cells	Direction and outcome
Class I HDAC inhibition (4-PBA)	as above, total 1.33 mM	as above	BubR1 effect shown with trichostatin A, not with 4-PBA; no 4-PBA concentration for the BubR1 endpoint [U]	not computable	HDAC2 and HDAC3 present in dividing cells	**CONFLICTED**, class excluded in Section 8.2. **E-UNRESOLVED, with a material AGAINST risk**
Heat-shock co-induction (arimoclomol)	label PK, highest studied exposure: Cmax 2,090 ng/mL = 6.66 uM; about 3.33 uM at the maximum labelled exposure [V]	**fu about 0.90, so free about 3 to 6 uM** [V]	no transferable concentration; PMID 15034571 is a SOD1-G93A mouse study with no in vitro value [V that none is given]	not computable	**NOT ESTABLISHED.** BUBR1 never shown to be an HSP70 client; co-induction requires pre-existing stress	UNKNOWN. **E-UNRESOLVED**
NAD+ raising, NAPRT route (niacin)	750 to 1,000 mg/day oral in the cited study [V]	endpoint is a metabolite pool, not a free drug concentration	North 2014 tested **NMN only**; no niacin `C_demo` exists [V]	not computable	**NOT ESTABLISHED.** "NAPRT is essential for NA to increase cellular NAD levels" [V], and NAPRT is liver-enriched (HPA nTPM 103.0 liver); fibroblast and marrow levels not established	UNKNOWN, with an inherited-direction risk. **E-UNRESOLVED**
NAD+ raising, NMN	human oral trial range 250 to 1,250 mg/day; no medicine label	plasma NMN after oral dosing **not established** [U] (the 0.135 uM figure used in the previous revision is nicotinamide, a different analyte)	**two values: 1 mM in MEFs; 500 mg/kg/day intraperitoneal in mouse** [V]	in vivo limb R = 0.10 to 0.51 on body-surface scaling; in vitro limb not computable	SIRT2 route intact in fibroblasts	TOWARD (in vivo, mouse) [V]. **E-UNRESOLVED** (frozen rule scores the less favourable limb)
NAD+ raising, NR	no medicine label	not established [U]	**no source tests NR on BubR1 protein** [V]	not computable	not applicable	UNKNOWN. **E-UNRESOLVED (no source for the arm)**

What this table establishes. For **five of seven arms the arithmetic cannot be done at all**, because no source establishes the concentration at which the proposed mechanism acts on BUBR1. One arm fails outright and is excluded. **The surviving mechanisms are therefore not ordered by engagement evidence, and this report does not order them.** E-UNRESOLVED removes nothing; it is a statement about the literature, not about a molecule.

Two further readings belong with it. The one arm whose achieved concentration is in the right order of

magnitude for its mechanism is 4-PBA (total plasma 1.33 mM, the range in which chemical chaperone and HDAC effects are usually demonstrated); it is also the arm carrying the aneuploidy risk in Section 8.2, and both facts belong together. And arimoclomol is **not** exposure-limited: it has the highest free concentration of any prescription molecule here. Its problem is the opposite: no demonstrated engagement step linking it to BUBR1 at any concentration, and a label that states verbatim that "the mechanism(s) by which arimoclomol exerts its clinical effects in patients with NPC is unknown".

Negative results from this test.

- **The NAD+ rise is not confined to NAD-deficient subjects.** The hypothesis under test was that niacin's effect requires a deficient baseline. PMID 32386566, verbatim: "Blood NAD+ increased in **all subjects**, up to 8-fold". The matched controls were not deficient and also rose. The 5.7-fold figure for those controls is **unverified**: it is not in the abstract, the full text is unreachable, and the paper carries an erratum we have not read (PMID 32640244), so the control rise is reported without a number. NOT CONFIRMED stands on the verbatim "all subjects" statement alone.
- **Body-surface scaling does not open a large gap for NMN.** The hypothesis under test was that the mouse in vivo dose is unreachable in humans. Human-equivalent dose = 500 mg/kg/day x (Km mouse 3 / Km human 37) = 40.5 mg/kg/day, that is about 2,432 mg/day for a 60 kg adult. Human oral trial exposures of 250 to 1,250 mg/day give R = 0.10 to 0.51, inside the ten-fold band. NOT CONFIRMED as a problem. The route mismatch (intraperitoneal in the mouse, oral in humans) is flagged and not silently ignored. The arm still scores E-UNRESOLVED only because the frozen rule scores the less favourable limb, and the in vitro limb cannot be computed.

7.2 Chemical-chaperone proteostasis, MECHANISM NOT TESTED (sodium phenylbutyrate, prescription US and EU)

Mechanism hypothesis. A chemical chaperone could raise the steady-state level of a destabilised missense BUBR1. 4-PBA "mediated a marked increase in secretion of functionally active alpha1-ATZ", a misfolded mutant protein (PMID 10677536, verified at abstract). **Grade G4.**

Mechanism status: NOT TESTED, downgraded from open. The previous revision carried this as an open hypothesis. The mechanism-class screen (Section 15) ran a fixed PubMed query, `(BubR1[tiab] OR BUB1B[tiab]) AND (chemical chaperone[tiab] OR phenylbutyrate[tiab] OR tauroursodeoxycholic[tiab] OR ambroxol[tiab] OR trehalose[tiab])`, and it returned **zero records**. So **no published source connects a chemical chaperone to BUBR1 at all**, and the rationale we carried is transferred wholesale from a different protein: PMID 10677536 concerns an ER-retained secretory glycoprotein with secretion as the endpoint, while BUBR1 is cytosolic and nuclear and never enters the secretory pathway. BUBR1 has never been shown to be a chemical-chaperone client.

Both of this molecule's arms are now problems. The chaperone arm has no source linking it to BUBR1, and no demonstrated concentration on BUBR1 can be taken from the source at all, because that paper is about alpha1-antitrypsin and never measures BUBR1 (Section 7.1). The other arm, class I HDAC inhibition, has plenty of sources and they point the wrong way: it is **excluded on direction** (Section 8.2), and at a total plasma concentration of 1.33 mM it is reached, so it cannot be dismissed as unattainable and the two activities are not separable at any reachable exposure. What survives is this: the molecule is approved, it is prescription in both regions with a paediatric label, and it is the one molecule in the whole screen whose achieved concentration sits in the order of magnitude where its proposed mechanism would act. That makes the untested question worth answering. It does not make the hypothesis supported, and we are not keeping it on a rationale the search could not find. If it works, it works in spite of one of its own mechanisms.

Regulatory facts, verified. Prescription in the US (PHEBURANE NDA216513, OLPRUVA NDA214860, generic tablets and powder; BUPHENYL discontinued; RELYVRIO discontinued) and authorised in the EU (Pheburane, 30 July 2013). Indication in both regions: adjunctive chronic management of urea cycle disorders, adult and paediatric; the label states that use of tablets is not recommended for neonates, infants

and children up to 20 kg. **The approved indication is nitrogen handling via phenylacetate and is unrelated to the chaperone activity**, so "prescription proteostasis modulator" must not be read as if the approved use were the proteostasis use.

Dependency. 1 flagged target, HDAC3, fraction 0.874, blood-lineage median -1.057. HDAC1 (0.040), HDAC2 (0.028) and HDAC8 (0.025) are not flagged.

Coverage gap, disclosed. This arm's class sits in the one slice of the mechanism-class screen's molecule universe, about 45 proteostasis and chaperone molecules, that was never evaluated drug by drug. So this arm carries a coverage gap in that screen on top of its evidence gap, and a molecule with a better chemical-chaperone case may exist in that slice unexamined.

Backing experiment. Steady-state BUBR1 level and cycloheximide-chase half-life as **separate** readouts in patient-derived cells, with a proteasome arm, a synchronisation or stage marker, and an interphase FISH or PCS-rate arm to detect the aneuploidy liability directly. **Falsifiers.** No change in steady-state level and no change in half-life; **or** any increase in PCS or missegregation rate, which fails the hypothesis regardless of what happens to BUBR1 level.

7.3 Heat-shock co-induction, MECHANISM NOT TESTED (arimoclomol, prescription US)

Mechanism hypothesis. Amplifying the heat-shock response could reduce misfolding-driven loss of a destabilised BUBR1, or support BUBR1 at the kinetochore. Arimoclomol is a co-inducer of heat shock proteins and delayed disease progression in a SOD1 mouse model (PMID 15034571, verified at abstract). **Grade G4.**

Mechanism status: NOT TESTED, and an adversarial citation check moved it back down from OPEN. The previous revision raised this class to MECHANISM OPEN on PMID 30496702 (Kakihana 2019, *FASEB J*). Read in full, that abstract does not support a BubR1-directed mechanism, and we report the reversal rather than keep the upgrade.

What the paper does say about BubR1, verbatim: "Knockdown of heat shock protein (Hsp)105 partially abrogated the mitotic delay with the loss of the kinetochore localization of BubR1 under heat shock conditions and accelerated mitotic progression under nonstressed conditions." Read precisely, the loss of kinetochore BubR1 is reported **under heat shock**; the nonstressed coordinate is a mitotic-timing phenotype and makes no claim about BubR1. The abstract also frames kinetochore BubR1 here as a heat-shock-induced, Mps1-dependent loading event, so there is no baseline pool for HSPH1 to hold in an unstressed cell.

And the paper's own proposed mechanism is not BubR1. Verbatim: HSPH1 knockdown "did not affect coprecipitation of Mad2 and BubR1 with Cdc20", and the conclusion is that HSPH1 "prevents abnormal cell division by contributing to SAC activation under heat shock and nonstressed conditions **by interacting with Cdc20 but not affecting formation of the mitotic checkpoint complex**". So the mechanism the paper proposes runs through Cdc20, it explicitly reports BubR1's association with Cdc20 as unaffected, and the only BubR1-specific observation in it is a correlative loss of kinetochore localisation under stress. The lagging-chromosome experiment has no stated condition in the abstract and the full text is unreachable, so it is marked **unverified** as to condition.

Verdict and the missing datum. No source in the searched record shows that raising chaperone levels raises BUBR1 protein level or restores BUBR1-dependent checkpoint function. The class search that produced this paper returned it as the only BubR1-linked record, and on reading it does not supply that datum. The arm is therefore **MECHANISM NOT TESTED**, not OPEN. The mechanism-class screen scored its class 14 as OPEN on the same source; our own reading after the fact-check is stricter, and we carry the arm at NOT TESTED and flag the divergence rather than take the more favourable of the two readings.

Three further problems, about the molecule rather than the class. It is a **co-inducer**, amplifying HSF1 only in already-stressed cells, and no stressor is proposed for unstressed patient fibroblasts; the cited

mouse model is a proteotoxically stressed system, so the stress-dependence clause of the pre-registration applies and first-step competence cannot be marked established. BUBR1 is not an established HSP70 or HSP90 client, and the premise that HSP70 with CHIP routinely triages destabilised clients **to** degradation is carried here as **unsourced**: no source in this report establishes it, and it is load-bearing, because it is the reason the sign of a general HSP70 rise is called unpredictable. Taken as unsourced, the sign is still not established in the other direction either, which is what keeps this arm at NOT TESTED. And the label states the mechanism of action is unknown. Note also that a class tag is not a direction: the same molecule universe contains quercetin, tagged to HSF1 and used in the literature as an HSF1 and HSP70 **inhibitor**, which moves this mechanism backwards.

Regulatory facts, verified. MIPLYFFA, NDA214927, approved 2024-09-20, prescription, indicated in combination with miglustat for neurological manifestations of Niemann-Pick disease type C in adult and paediatric patients 2 years and older. **EU: not authorised.** Two applications must not be conflated: Miplyffa (EMEA/H/C/005203, Orphazyme) was withdrawn on 22 March 2022; Meplyffa (EMEA/H/C/006736, Zevra Denmark) received a negative CHMP opinion on 23 July 2026 with a re-examination requested, which was on the CHMP October 2026 agenda and had no final outcome as of 2026-10-20. Reported as a negative opinion under re-examination, never as a refusal.

Dependency. 0 flagged targets **over 1 of 3 originally pre-registered genes measured**: HSF1 is measured at fraction 0.168 (below the flag, not negligible); HSPA1A and HSPA1B have no DepMap column. A zero flag count over one measured gene of three is not comparable to a zero over five of five, and the previous revision's tie-break rewarded exactly that. The tie-break is withdrawn with the ordinal list. The mechanism-class screen measured the rest of the route: HSPH1 0.002, HSPA4 0.010, HSPB1 0.000, HSP90AA1 0.052 and HSP90AB1 0.124, none flagged, while the constitutive chaperones HSPA8 (0.864) and HSPA5 (0.993) are. The gene this class's rationale rests on, HSPH1, carries no dependency signal at all. **Coverage gap, disclosed:** like the arm above, this class sits in the proteostasis and chaperone slice of the molecule universe that was never evaluated drug by drug, so a better-supported co-inducer may exist in it unexamined.

Backing experiment, which is now the experiment that would supply the missing datum rather than test an established rationale. Reproduce the HSPH1 knockdown readout of PMID 30496702 as the assay control, then ask two questions the paper leaves open: **does a heat-shock co-inducer raise HSPH1, BUBR1 kinetochore occupancy and BUBR1 steady-state level in UNSTRESSED human cells, and is any effect on the checkpoint BubR1-dependent rather than Cdc20-dependent?** Run paired stressed and unstressed arms in patient-derived cells, with HSP70 and HSPH1 induction measured in the same lysates as the engagement control, mitotic checkpoint complex composition measured alongside, and the missegregation rate scored. **Falsifiers.** No HSPH1 or HSP70 induction at a reached exposure in unstressed cells (the mechanism is not engaged where it would have to work); or induction with no change in BUBR1 kinetochore occupancy or steady-state level (BUBR1 does not benefit); or an increase in PCS or lagging-chromosome rate.

7.4 NAD+ raising into a SIRT2-dependent step, MECHANISM OPEN (niacin, prescription US; NMN and NR, dietary supplements)

Mechanism hypothesis. Raising NAD+ raises SIRT2 activity, which deacetylates BubR1 at K668 and slows its degradation (PMID 24825348, Fig 6F and 6G). Independent pharmacological support for the axis, in the opposite direction and with a tool compound: a selective SIRT2 inhibitor destabilises BubR1 (PMID 25672491, Section 4.5). Contested by PMID 25285631 (Section 4.4). **Grade G4 for niacin and NR; NMN is G3-contested, which is not one of the four defined grades and is recorded as a contested label rather than a grade (Appendix A).**

Mechanism status: OPEN, with four stated problems.

1. **The first step is not established in the relevant cell type.** "NAPRT is essential for NA to increase cellular NAD levels" (PMID 17604275, verbatim), and NAPRT is liver-enriched in the Human Protein Atlas. Whether fibroblasts or marrow express it is a cheap, untested question and is named as a required test. Absence of data is not read as absence of the enzyme.
2. **The anchor paper tested NMN only.** No source tests niacin or NR on BubR1 protein. For NR this is an absence of evidence, not evidence of absence.
3. **An inherited direction problem, now stated explicitly.** The previous revision removed nicotinamide on criterion C4 because nicotinamide inhibits SIRT2 at a plasma concentration humans reach. Our own niacin source infers that niacin "was likely metabolized as NAM via salvage pathway instead of Preiss-Handler pathway" (PMID 32386566). If niacin acts through nicotinamide, it inherits the direction problem that removed a sibling molecule. The previous revision recorded this as C4-UNRESOLVED without saying it was the same question. It is the same question. Separately, oral NMN and NR are reported to be partly converted to nicotinamide in vivo; that report is carried here as **unsourced**, because no identifier for it was resolved for this revision, and the conversion has not been checked for either molecule.
4. **C4 itself cannot settle any of this.** See Section 9.3. The criterion is non-discriminating, so the unresolved reading is reinforced rather than resolved.

A second, independent route into this class, and it is the cheapest open question in the report. Every doubt above attaches to SIRT2: the K668 against K250 dispute, the single-laboratory dependence of the K668 result, and our own DepMap null for SIRT2 against BUB1B. The mechanism-class screen found a SIRT1 transcriptional route that inherits none of them: SIRT1 is reported to bind the BubR1 promoter and upregulate its expression, from a group independent of the K668 author line (PMID 35639235, verified at abstract and record level). It is also the single most consequential **unverified** claim in that screen, and the reasons are stated rather than smoothed over: the full text is unreachable on every route available to this work, the SIRT1 manipulation was genetic rather than pharmacological, whether the effect is on transcript or on protein is **not established**, and six records pair SIRT1 with *BUB1B* while only this one makes the promoter claim, with no corroboration and no contradiction. It changes no verdict here. The deciding experiment is cheap and is stated in Appendix B class 10.

Also unanswered: whether there is a deficit to correct. Both anchors for this family are deficiency-correction settings (aged mice with declining NAD+; patients with documented systemic NAD+ deficiency measured from a deficient baseline). Nobody has asked whether NAD+ is low in this disease, nobody proposes measuring it, and no headroom argument exists. The statement that whole-blood NAD+ is dominated by anucleate erythrocytes that never divide is carried here as **unsourced**, and if it holds a whole-blood rise is a weak proxy for anything happening in a dividing cell.

Regulatory facts, verified. *Niacin:* every single-ingredient oral niacin product in Drugs@FDA is Prescription or Discontinued; none is over the counter. Active prescription products include NIACOR immediate-release (ANDA040378) and extended-release generics (ANDA204934, ANDA090892, ANDA200484, ANDA201273); the NIASPAN brand (NDA020381) is Discontinued. US indication: reduction of elevated lipids. **EU: not authorised**; the only EU centrally authorised niacin lipid product, nicotinic acid with laropiprant, was suspended in January 2013 and withdrawn. Non-prescription niacin is sold as a dietary supplement, which is not a drug approval. Paediatric: the prescription labels state use is not established at 16 years and under. *NMN and NR:* neither is an approved medicine. US: FDA's response in docket FDA-2023-P-0872 (document FDA-2023-P-0872-2754, 29 September 2025) finds NMN not excluded from the dietary supplement definition, reversing the 2022 position; NR holds GRAS notice GRN 635 (closed 3 August 2016, "no questions") and NDI notifications 882 and 1062. EU: NR chloride is authorised as a novel food (Commission Implementing Regulation (EU) 2020/16); NMN is absent from the Union list.

Dependency. Niacin 0 flagged of 5 measured (NAPRT 0.012, NMNAT1 0.188, NMNAT2/3 and precursor

entry steps 0.000 to 0.002, SIRT1/SIRT2 0.004/0.006). NMN and NR 0 flagged. NAMPT, the entry enzyme for the nicotinamide route, is flagged at 0.296 with a blood-lineage median of -0.936, and NMNAT1's blood-lineage median is -0.911 although it is not flagged by the all-lines rule. Those two numbers are why normal marrow is the named next safety test for any NAD-raising mechanism.

Backing experiment. NAPRT and NMRK1/2 expression in the target cell type first, as a go/no-go on the first step; then steady-state BUBR1 level and half-life at an NAD+ exposure matched to a measured intracellular NAD+ rise, not to a plasma figure, with SIRT2 inhibition as the mechanism control and intracellular nicotinamide measured in the same cells. **Falsifiers.** No intracellular NAD+ rise in the target cell type (the first step is absent); or an intracellular NAD+ rise with no BUBR1 change (the lever does not act); or an intracellular nicotinamide rise into the inhibitory range (the direction inverts); or an increase in PCS rate. **Human context.** Registered NMN and NR trials exist in sarcopenia, frailty and kidney disease (Section 9, Table 5). **None in *BUB1B*, SIRT2, MVA or aneuploidy.**

The closest published human evidence to this population, and what it does and does not report. One trial sits nearer to the population this report is about than anything else we found: a randomised pilot of 20 adult survivors of childhood cancer at risk for diabetes, median age 35.5 years, given nicotinamide riboside (PMID 39387327, verified at abstract). Its proximity is the reason for naming it, and its limits are the reason it changes nothing. **The abstract reports feasibility and adverse events only, and no efficacy signal is claimed here or available from it.** The participants are adults, not children; the endpoints are metabolic, not BUBR1, checkpoint or aneuploidy endpoints; and a cancer-survivor cohort is not an MVA cohort, although the shared feature of childhood cancer is why it is the nearest population on record. It establishes that an NAD+ precursor has been administered in a randomised design to survivors of childhood cancer and that the design was feasible. It establishes nothing about this mechanism, and Section 11.4's unanswered question, whether raising NAD+ supplies a substrate advantage to a transformed clone, is not touched by it.

A fourth agent of this class was evaluated and is not carried: acipimox. Acipimox is a nicotinic acid derivative that its own literature describes as an NAD+ precursor, so it belongs to this arm's mechanism class rather than opening a new one. It is declined on measurement, not on judgement. The step this arm needs was tested directly in human tissue and did not occur: in eleven older adults with probable sarcopenia given acipimox for four weeks, the pre-specified primary outcome was skeletal muscle NAD concentration, the median change was -0.003 umol/g (IQR -0.058 to 0.210, p = 0.26), and the authors state that acipimox did not improve skeletal muscle NAD concentration, biochemical markers or physical function (McDonald 2025, PMID 40100528). The NAD+ effect that gives the molecule its label sits one system away from that result: the human crossover trial in twenty-one people with type 2 diabetes reports increased ex vivo mitochondrial respiration, and the direct effect on NAD+ levels is confirmed in C2C12 myotubes, a mouse cell line, rather than in the human muscle of the same trial (van de Weijer 2015, PMID 25352640).

Why niacin stays in this arm and acipimox does not. The difference the evidence supports is between a first step that is untested and a first step that was tested and returned null. Problem 1 above states that whether the target cell type can use nicotinic acid at all is untested for niacin, and that absence of data is not read as absence of the enzyme. For acipimox the equivalent measurement exists in human muscle and is null. No stronger claim is made here: this is not evidence that niacin raises NAD+ in the relevant cell type, which remains the untested question that gates the whole arm.

What this adds to Section 9.3. The pattern is the one already recorded there for NR and NMN, where the mechanism is demonstrated in a non-human or non-target system and the direct measurement in human tissue is null. Acipimox makes three agents of this class failing at the same step, measured three different ways, which is a result about the class and therefore about this arm. An arm that keeps a mechanism OPEN should carry the strongest evidence against its own first step, and this is it.

Regulatory class, and why the universe screen did not reach it. C1 returns no authorised drug product in the United States: Drugs@FDA and the openFDA label endpoint return nothing for acipimox or

for Olbetam, with niacin returning eight products as the positive control for the same query. The EMA reading is not reported, because the same query returns nothing for a medicine that is centrally authorised and therefore cannot separate absence from a failed query. Neither arm of the universe screen reached this molecule: arm A screens curated drug-to-gene interactions for the named targets and acipimox acts upstream of them rather than on them, and arm B indexes titles and abstracts. Both limits are ones that pre-registration states for itself in its Section 2.3.

7.5 Research-stage ideas, not evaluated as medicines

- **Transient BUBR1 mRNA delivery.** Ectopic BUBR1 cDNA restored checkpoint activity in MVA patient cells and overexpression of low-abundance mutants to wild-type levels fully restored the nocodazole response (PMID 20516114, Fig 4D, verified). The delivery in that paper was plasmid cDNA; an mRNA pulse is our own proposal, not the paper's. It is the only idea here that targets the effect size Section 4.3 actually demonstrates. Suggested test: a single-dose mRNA pulse in patient-derived fibroblasts, read out by mitotic index and interphase FISH.
- **MAD2 and insulin-receptor endocytosis.** MAD2 binds the insulin receptor and facilitates BUBR1-dependent AP2 recruitment (PMID 27374329, verified). Untested here. Suggested test: insulin-uptake assay on patient lymphoblastoid cells with BUBR1 or MAD2 knockdown controls.
- **Gene-corrected blood stem cells engrafted without conditioning. Research stage, outside the approved-medicine scope, and not carried forward.** This was examined as a delivery route for the gene-level correction of class 21, because it would avoid the conditioning regimen that a cancer-predisposition phenotype makes hard to justify. Gene-corrected haematopoietic stem cells have been reported to engraft progressively in non-conditioned patients with Fanconi anaemia (PMID 31501599, cited for that qualitative finding only; cohort size not confirmed from the abstract), with an open-label phase of the same programme reported subsequently (FANCOLEN-1, PMID 39642902), and postnatal trafficking has been used to enable in vivo haematopoietic stem cell gene therapy in a mouse (PMID 40437086). **Two recorded findings are why it is not carried forward, and they are the point of recording it at all.** The first is a precondition: conditioning-free engraftment in those reports depends on the uncorrected stem cells being at a competitive disadvantage, so that corrected cells outgrow them unaided. In Fanconi anaemia that disadvantage is a feature of the disease. No source establishes it for *BUB1B*-mutant haematopoiesis, and until it is established the mechanism that makes the approach work is absent. The second is reach: the compartment is blood, and MVA's growth, neurological and solid-tumour burden is not confined to the blood compartment, so even a fully corrected marrow would leave the phenotype this report names in Section 2 largely untouched. Labelled, as class 21 is, research stage and outside the approved-medicine scope; stated as a modality examined and set aside on a named precondition and a named limit of reach, not as a recommendation and not as a ranked option.

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8. Mechanism exclusions

Each line names the test that decided it and what that test rules out. Exclusions are about mechanisms and sources, not about molecules being unsafe or about anyone's treatment.

8.1 PP2A activation: EXCLUDED on direction, using our own cited source

Three findings, all verified verbatim at source in `T2-EXPOSURE-01`.

(a) The circulating species inhibits PP2A. PMID 23926298, verbatim (the paper carries an erratum, *Blood* 2014;123(19):3056, which we have not read): "Jak2(V617F) also utilizes an alternative sphingosine kinase-1-mediated pathway to inhibit PP2A and that FTY720-P, acting as a sphingosine-1-phosphate-receptor-1 agonist, elicits signals leading to the Jak2-PI-3Kgamma-PKC-SET-mediated PP2A **inhibition**."

(b) The activating action belongs to the species that is least available. Same paper, verbatim: "FTY720 antileukemic activity requires **neither** FTY720 phosphorylation (FTY720-P) nor SET dimerization or

ceramide induction but depends on interaction with SET K209." The label states parent fingolimod is 23.3% of blood AUC, is 86% sequestered in red blood cells and is over 99.7% protein bound. The arm needs the fraction of a fraction of a fraction, and the exposure gap is not closable: the label states that higher exposures give "a greater incidence of adverse reactions without additional benefit".

(c) Even successful PP2A activation points the wrong way. PP2A-B56 is recruited **by** BUBR1 (PMID 33207204, PMID 23079597, PMID 23789096, Section 4.6), so in a BUBR1-deficient cell the limiting quantity is the scaffold, not phosphatase activity. At the kinetochore the PP2A-B56 relay extinguishes the checkpoint. Driving the silencing step in a cell that already fails to hold the checkpoint is scored AGAINST under the frozen node rule.

Pre-registered falsifier, checked: a source showing FTY720-phosphate activating PP2A, or kinetochore PP2A-B56 strengthening the checkpoint. Neither exists in the searched record; both sources say the opposite. The exclusion stands. **It is a removal on mechanism direction and exposure, not on safety.** The DepMap flags that previously demoted this candidate (PPP2CA 0.826, PPP2R1A 0.779) are retained in Table 2 as measurements and play no part in the exclusion, which is the correct use of them: a knockout is a loss-of-function proxy and the proposal was an activator, so the knockout direction was always opposite to the drug's.

8.2 Class I HDAC inhibition: EXCLUDED on direction, class-level, zero entrants

This was the most promising-looking mechanism available to us, and it is the one we tested hardest.

The lead. PMID 28985013 (Park et al 2017, *FEBS J*), verbatim: "Histone deacetylase (HDAC) 2 and 3 bound to acetylated BubR1 exclusively in mitosis and led to the polyubiquitination of BubR1. Subsequent degradation of BubR1 resulted in the disassembly of the mitotic checkpoint complex." Inhibiting HDAC2 and HDAC3 therefore predicts BubR1 stabilisation, which is the exact effect this report is looking for, and it is the only BubR1-specific, published, drug-targetable abundance mechanism we found. It is carried here as the lead that motivated the test.

Two things in the lead that the one-line version of it does not carry. The process it describes is the normal, required termination of checkpoint signalling at mitotic exit, not a pathological route of BubR1 loss: the paper's own framing is that "BubR1 deacetylation is a signal that initiates mitotic exit" and "a cue for SAC silencing", and its stated consequence of blocking it is that "sustained BubR1 acetylation arrests the cells in metaphase, although chromosome congression is achieved", which is a mitotic block, not restored fidelity. And the paper's only HDAC-inhibitor arm is a drug-resistance experiment in BRCA2-dysfunctional cells, not a BubR1-stabilisation experiment. **No experiment in that record shows BubR1 protein rising, or a checkpoint working better, when HDAC2 and HDAC3 are inhibited.**

The screen. We pre-registered a search of every approved class I HDAC inhibitor against regulatory status, dependency, exposure, direction and grade, with the rescuing falsifier fixed in advance. The universe surfaced no approved class I HDAC inhibitor beyond the seven named at pre-registration. **Zero drugs entered the candidate list.** Valproate reaches its HDAC-inhibitory concentration at the upper therapeutic range and is excluded on direction; vorinostat, romidepsin, belinostat and givinostat are excluded on direction, with romidepsin, belinostat and givinostat also carrying labelled myelosuppression; panobinostat failed the approval criterion (no retrievable FDA label or drugsfda record on 2026-10-20, recorded as unverified against a withdrawal notice). Givinostat matters as a declared expected prior rather than as a discovery: it carries a chronic paediatric indication in Duchenne muscular dystrophy for patients 6 years and older, so the class cannot be dismissed as having no paediatric-compatible member. One exists, is dosed chronically in children, and is still excluded, on direction.

The direction evidence, six primary sources, all verified by direct record retrieval.

PMID	Verbatim finding	Bearing
PMID 15753652	Trichostatin A "abrogated the phosphorylation and kinetochore localization of the mitotic checkpoint protein BubR1" **after paclitaxel and nocodazole treatment**; separately, TSA alone "led to missegregation of chromosomes"	The one experiment in the searched record with BubR1 as an endpoint under HDAC inhibition, and it points the wrong way. The BubR1 result is conditioned on spindle poison; the missegregation result is TSA alone
PMID 17616695	"HDAC inhibition renders the SAC ineffective"; premature sister chromatid separation in cells with the checkpoint already activated; "This effect was transcription-independent"	Checkpoint override, not a gene-expression artefact
PMID 17828304	HDAC inhibitors "induce aberrant mitosis and can overcome the spindle assembly checkpoint"; passenger-protein concentrations "insufficient to regulate the phosphorylation of the kinetochore checkpoint component BubR1"; mitotic slippage	Checkpoint override through failed BubR1 regulation; the counter-evidence to the lead
PMID 20538840	A "class I-selective histone deacetylase inhibitor" "shared the ability of the pan-isoform HDACi to trigger defective mitosis and promote mitotic slippage"	Class I selectivity does not escape the problem, which removes the obvious rescue. **Confounded, and we say so:** the paper's headline finding is that the same agent also destabilises microtubules, so this is not a clean isolation of HDAC inhibition
PMID 18832068	"HDAC3 knockdown induced spindle assembly checkpoint **activation** and sister chromatid dissociation"; HDAC3 "required for Shugoshin localization at centromeres"	Cited only for the cohesion requirement at HDAC3, the lead's own target. **It reports checkpoint activation, the opposite direction from the override papers above, and is not stacked with them**
PMID 18326024	HDAC3 knockdown gave "a collapsed mitotic spindle"; "kinetochore-microtubule attachments were impaired"; the same with trichostatin A "independent of transcriptional regulation"	Cited only for the attachment requirement at HDAC3, the process already failing here. **Its own abstract states that "the kinetochores and the spindle assembly checkpoint appeared intact in HDAC3-deficient cells", so it must not be used to argue that HDAC inhibition disables the checkpoint**
PMID 16857700	Human TK6 lymphoblasts: "Significant levels of aneuploidy at the 200nM TSA dose"; aneuploidy "via a nondisjunction event"; "We conclude that TSA is both a clastogen and aneugen"	**HDAC inhibition produces aneuploidy in human cells**, the phenotype the hypothesis exists to reduce

What each source is and is not doing in that table. The exclusion rests on three papers: PMID 15753652 (BubR1 phosphorylation and kinetochore localisation abolished), PMID 16857700 (aneuploidy in human lymphoblasts, by nondisjunction rather than lagging chromosomes) and PMID 17616695 (transcription-independent checkpoint override). The other three are cited narrowly and each carries a stated limit: PMID 20538840 shows that class I selectivity does not rescue the class but is confounded by its own headline microtubule-destabilising finding; PMID 18832068 is cited only for HDAC3's cohesion requirement and in fact reports checkpoint **activation**, the opposite direction from the override papers; and PMID 18326024 is cited only for HDAC3's attachment requirement, since its own abstract reports the kinetochores and the checkpoint as intact in HDAC3-deficient cells. The class verdict does not depend on any of those three.

Pre-registered falsifier F1, checked: a primary source showing increased BubR1 protein and a functioning checkpoint after class I HDAC inhibition at or below an achievable free plasma concentration, uncontradicted. Of the records returned for `(BubR1 OR BUB1B) AND (HDAC OR histone deacetylase)` at the frozen retrieval on 2026-10-20, which was 14, exactly one pairs an HDAC inhibitor with BubR1 as a protein endpoint, and it is PMID 15753652. **F1 not met at that retrieval. The count is not stable and we say so rather than printing an unreproducible figure:** re-running the same query string on the same date returns 25 records unfielded and 11 with `[tiab]` field tags, so the result depends on field scoping. The additional records have not been adjudicated, so F1 is reported as not met over the 14 retrieved at the run, and re-adjudication over the wider set is an open item. The verdict is class-level: reproduced across trichostatin A, pan-isoform and class I-selective agents, and transcription-independent in two sources.

DepMap cross-check (method unchanged from `T2-P5`). Of 11 HDAC genes, HDAC3 is the only flagged isoform (fraction 0.874, blood-lineage median -1.057), by a wide margin, and it is the lead's own named target. HDAC1 (0.040), HDAC2 (0.028) and HDAC8 (0.025) reproduce `T2-P5` to three decimals.

Consequence for an open hypothesis. Sodium phenylbutyrate's class I HDAC activity (HDAC1, HDAC2, HDAC3, HDAC8) is covered by this class-level verdict. For a hypothesis whose aim is a better-functioning checkpoint, that component **points the wrong way: it is a liability to be argued around, not a second mechanism in its favour**. Its total plasma Cmax of 1.33 mM sits in the same millimolar order as published phenylbutyrate HDAC-inhibitory concentrations, so the activity cannot be dismissed as unreachable, although by the pre-registered rule the free concentration is UNRESOLVED and no "reaches" verdict is recorded.

Limits of this exclusion. The counter-evidence is largely trichostatin A and oncology-grade inhibitors in cancer cell lines at concentrations chosen to block HDACs completely. None of it is in *BUB1B*-mutant or patient-derived cells, and none in a child. It establishes the direction of the class, not a quantitative prediction.

8.3 Other mechanism exclusions

Mechanism or molecule	Deciding test	Result	What it rules out
Nicotinamide as a molecule on this mechanism	`T2-CLASS-01`, criteria C1 and C4	No FDA-approved oral single-agent product exists, so its class is dietary supplement, not OTC drug; and C4 read "opposes": a SIRT2 IC50 of 70 +/- 5 uM reported in the text of PMID 30986062, **where the paper's Table 1 column order is internally inconsistent between its nicotinamide and EX527 rows, so the prose assignment to SIRT2 is the only self-consistent reading** (verified in the free full text at PMC6901289), against a mean peak plasma level of 1.28 mM at oral doses up to 6 g (PMID 8532910, pooled across the 3, 4, 5 and 6 g arms rather than measured at 6 g alone), with the hypothesis's own source reporting that nicotinamide treatment **reduced** BubR1 protein (PMID 24825348, Fig 1A and 1B)	Treating niacin and nicotinamide as one candidate with one class. **It does not rule out nicotinamide on safety, and after Section 9.3 the C4 half of this entry is not evidence of harm.** The 6 g exposure is above the human dose window (Section 9.3)
mTOR inhibition (everolimus, sirolimus)	`T2-P4`	MTOR flagged, fraction 0.986, blood median -1.130	The mTOR route as a candidate target class, on the pre-registered dependency flag
APC/C inhibition	`T2-P4`	Every subunit measured is flagged, most near the ceiling: CDC20 1.000, ANAPC1 1.000, ANAPC5 1.000, CDC16 1.000, CDC23 1.000, CDC27 1.000, ANAPC11 0.991, ANAPC2 0.977, ANAPC4 0.994, UBE2C 0.767, UBE2S 0.292	The APC/C route as a candidate target class, **on the dependency rule alone**. Its direction evidence points the other way: "APC/C-Cdc20 was responsible for BubR1 degradation during mitosis" (PMID 19407811), so inhibiting it should raise BubR1. See the disclosure below
PLK4 inhibition	`T2-P4`	PLK4 flagged, 0.982	The PLK4 route as a candidate target class

Mechanism or molecule	Deciding test	Result	What it rules out
KIF18A inhibition	T2-P4	KIF18A flagged, 0.885	KIF18A as a candidate target
The SIRT2 shared-stability route at fitness level	T2-DEPMAP-01	SIRT2 versus BUB1B Spearman rho +0.043, p = 0.14, against a pre-stated prediction of positive correlation; 5 of 5 positive controls passed (Section 9.4)	A shared fitness-level stability route in cancer lines, at the pre-stated threshold. **It does not test the acetylation mechanism, and it does not touch the pharmacological result in PMID 25672491**
Any pre-specified combination lever gene (23 candidates)	PRE-REG-B	No gene of 17,916 meets rho >= 0.20 and genome percentile >= 95	**Relabelled (Section 9.4): this negative is a property of the rule, not of the biology.** The threshold excludes the screen's own BUB1 positive control at +0.192
Read-through agents for the truncating allele	Reasoned exclusion, not a test	A premature stop makes read-through **mechanistically in scope**. It is set aside because a predicted NMD-null transcript leaves little template to read through, and read-through efficiency is strongly context-dependent across drugs and stop codons (PMID 39174735, verified at abstract)	Nothing. This is an exclusion by reasoning and is labelled as such. An earlier draft wrongly called it inapplicable to the genotype class; that claim is withdrawn. The mechanism-class screen reaches the same place by two different routes and records the class as MECHANISM NOT TESTED (Section 15, class 19): there is currently **no approved designed read-through medicine** (the one such product's EU conditional authorisation was not renewed on 28 March 2025 and has expired, and openFDA returns nothing on either endpoint), and for the aminoglycoside limb, which is approved, the literature search returns count 0

A disclosed asymmetry in our own exclusions. APC/C inhibition is excluded above on the dependency rule, while **proteasome inhibition carries the same objection at the same magnitude and is not excluded** (Section 15, class 12: PSMB1 0.997, PSMB2 1.000, PSMB5 0.947, all flagged). The reason is procedural, not scientific: the frozen pre-registration of the mechanism-class screen forbids extending the T2-P4 precedent to classes T2-P4 did not name, and T2-P4 named the APC/C subunits. A dependency flag is also not a direction test, and in the APC/C case the direction evidence points the other way (PMID 19407811, "APC/C-Cdc20 was responsible for BubR1 degradation during mitosis"). So one class is excluded by a recorded decision of ours and a second class with the same liability is left open. Readers should weigh the APC/C exclusion accordingly: it is our rule doing the work, not the biology.

The standard of evidence behind that asymmetry, applied symmetrically. Both of those classes rest on a laboratory tool compound rather than a medicine: trichostatin A carried the class-level exclusion of the HDAC route in Section 8.2, and MG-132 carries the one supporting direction result for the proteasome route. The rule we apply is the same in both places: tool-compound evidence is admissible for a verdict about a **mechanism class**, and never sufficient for a verdict about a **specific medicine**. That is why the HDAC class is excluded at class level while the approved proteasome inhibitors are individually NOT EVALUABLE, which means no source measures them on BUBR1, not that they were examined and cleared.

9. Evidence we generated

All DepMap analyses used the public 24Q4 CRISPRGeneEffect and Model files (1,178 lines; 348 blood-lineage lines, 135 with gene effects). Cancer lines only, never normal marrow. No patient data was used in or sent to any of these analyses.

Table 2. On-target dependency of candidate and comparator genes (`T2-P4`, `T2-P5`, and the index gene added in R2-E). Flag rule, fixed before the data were opened: fraction of lines with Chronos gene effect below -0.5 at or above 0.25.

Gene	Role here	Fraction < -0.5	Flag	Blood-lineage median
BUB1B	**the index gene itself**	**0.973**	**YES**	**median effect -1.031; SD 0.307, at the 95.4th percentile of all 17,916 genes**
CDC20	excluded route	1.000	YES	-1.894
ANAPC4	excluded route	0.994	YES	-1.132
MTOR	excluded route	0.986	YES	-1.130
PLK4	excluded route	0.982	YES	-1.281
KIF18A	excluded route	0.885	YES	-0.839
HDAC3	4-PBA; excluded HDAC route	0.874	YES	-1.057
PPP2CA	excluded PP2A route	0.826	YES	-1.166
PPP2R1A	excluded PP2A route	0.779	YES	-0.510
UBE2C	excluded route	0.767	YES	-0.572
NAMPT	nicotinamide route (excluded molecule)	0.296	YES	-0.936
NMNAT1	shared NAD route	0.188	no	-0.911
HSF1	arimoclomol	0.168	no	-0.164
HDAC1 / HDAC2 / HDAC8	4-PBA	0.040 / 0.028 / 0.025	no	-0.196 / -0.030 / -0.118
NAPRT	niacin	0.012	no	-0.139
SIRT1 / SIRT2	NAD effector step	0.004 / 0.006	no	+0.089 / -0.036
SPHK2 / SET	excluded PP2A route	0.001 / 0.009	no	-0.169 / -0.149
S1PR1 / S1PR3 / S1PR4 / S1PR5	excluded PP2A route	0.002 / 0.003 / 0.002 / 0.000	no	+0.007 / -0.009 / -0.048 / -0.030
NMNAT2 / NMNAT3 / NMRK1 / NMRK2 / SLC12A8 / HCAR2	precursor entry steps	0.000 to 0.002	no	-0.027 to +0.179
ABL1 / SRC	pre-set comparator genes, not a candidate target	0.011 / 0.041	no	-0.041 / -0.129
HSPA1A / HSPA1B	arimoclomol	not measured	not measured	not measured

Reading. BUB1B's own dependency was never reported in the previous revision and is the number a reader needs to interpret the nulls below; it is added here. BUB1B is broadly essential (97.3% of lines below -0.5) **and** among the 5% most variable gene-effect vectors in the dataset, which refutes the objection that a correlation against it is uninformative because its variance is compressed. NAMPT meets the flag (0.296) and is also essential in blood lineages. NMNAT1 is not flagged by the all-lines rule but has a strong blood-lineage median (-0.911), so the flag rule under-counts it; reported as such, not re-flagged. ABL1 and SRC were pre-set comparators and are not targeted by any mechanism here. `T2-P5` reproduced the three overlapping `T2-P4` values (NAMPT, NMNAT1, SIRT2) exactly in an independent run, and the HDAC screen reproduced HDAC1, HDAC2, HDAC3 and HDAC8 to three decimals in a third.

Table 2b. Dependency of the mechanism-class targets measured by the class screen (Section 15). Same flag rule, same files, same method. 60 targets measured, all sixty returning a column; seven re-measured controls reproduced the earlier runs exactly (SIRT2 0.006, HDAC3 0.874, MTOR 0.986, CDC20

1.000, PLK4 0.982, KIF18A 0.885, TTK 0.948). One gene, SMG1, has a reduced denominator and says so in its own row. Listed descending by fraction.

Gene	Mechanism class	Fraction < -0.5	Flag	Note
AURKB	other mitotic targets	1.000	YES	blood median -2.420. Not the single most essential gene on either metric: it ties at 1.000 with CDC20, ANAPC1, ANAPC5, CDC16, CDC23, CDC27 and PSMB2, and PLK1 (-2.772) and CDC27 (-2.632) have lower blood-lineage medians
PSMB2	proteasome inhibition	1.000	YES	blood median -2.050
PLK1	other mitotic targets	0.999	YES	blood median -2.772
UPF1 / UPF2	NMD inhibition	0.998 / 0.997	YES	premature-termination-codon rescue route
PSMB1	proteasome inhibition	0.997	YES	blood median -1.804
TOP2A / TOP1	topoisomerase cytotoxics	0.997 / 0.979	YES	class excluded on direction
TUBA1B / TUBB	microtubule-targeting agents	0.997 / 0.964	YES	class excluded on direction
HSPA5	constitutive chaperone	0.993	YES	not a candidate target; contrast with HSPH1 below
TTK	TTK/MPS1 inhibition	0.948	YES	class excluded on direction
PSMB5	proteasome inhibition	0.947	YES	blood median -0.939
SMG1	NMD inhibition	0.946	YES	**Reduced denominator, disclosed.** SMG1's fraction is over **349 lines, not 1,178**, and its blood-lineage median of -1.132 is over **22 blood lines, not 135**. The fraction is taken over lines with a value, as the frozen method specifies. **SMG1 is the only gene in the 60-row panel for which either denominator differs**, verified by checking every row of the source table `nc_depmap2.tsv`; the only other exceptions anywhere in the project are HSPA1A and HSPA1B, reported as not measured
HSPA8	constitutive chaperone	0.864	YES	not a candidate target
BCL2L1	senolytic clearance	0.846	YES	BCL2 0.028 and BCL2L2 0.019 are not flagged
EP300	lysine acetyltransferase inhibition	0.284	YES	blood median -0.761; CREBBP 0.076 and KAT2B 0.000 are not flagged. Class excluded on direction
TUBB4B	microtubule-targeting agents	0.262	YES	
FOXM1	transcriptional upregulation	0.206	no	
HSP90AB1 / HSP90AA1	HSP90 inhibition; arimoclomol route	0.124 / 0.052	no	the obstacle in this class is evidence, not dependency
SIRT7 / SIRT3 / SIRT6	sirtuin route	0.033 / 0.003 / 0.000	no	no sirtuin isoform is flagged
HSPA4 / HSPB1 / HSPH1	chaperone co-induction	0.010 / 0.000 / 0.002	no	**HSPH1, the gene this class's rationale rests on, carries no dependency signal**
PPP2R5A to PPP2R5E	PP2A-B56 subunits	0.000 to 0.036	no	the flags in the PP2A route sit on the catalytic and scaffold subunits, not on the B56 subunits that do the BUBR1-directed work
ATG5 / ATG7 / ULK1	autophagy modulation	0.001 / 0.001 / 0.000	no	

Table 3. Flagged-target count per mechanism arm, with its denominator. The previous revision counted flags without a denominator, which rewarded having fewer measurable targets. The denominator

is now shown and the count no longer orders anything.

Arm	Flagged	Flagged genes	Genes measured / pre-registered	Not measured
Niacin	0	none	5 / 5	none
NMN	0	none	5 / 5	none
NR	0	none	5 / 5	none
Arimoclomol	0	none	**1 / 3**	HSPA1A, HSPA1B
4-PBA	1	HDAC3	4 / 4	none
Fingolimod (excluded)	2	PPP2CA, PPP2R1A	8 / 8	none
Nicotinamide (excluded)	1	NAMPT	1 / 1	none

Table 4. Negative results, reported in full.

Test	Pre-registered rule	Result	What it rules out
`T2-DEPMAP-01`, SIRT2 versus BUB1B	Positive correlation predicted if SIRT2 acts through BUBR1 stability; falsified if rho is near zero	rho +0.043, p = 0.14. **Not relabelled** after the power test (Section 9.4)	A shared fitness-level stability route in cancer lines. Not the acetylation mechanism, and not the pharmacological result in PMID 25672491. Honest limit: power against an effect the size of the one observed is only 0.341
`PRE-REG-B`, genome-wide lever screen (17,916 genes)	Co-dependent requires rho >= 0.20 **and** genome percentile >= 95	No gene qualifies. PPP2R5B +0.098 at percentile 96.5 is the strongest. **Relabelled** (Section 9.4)	Nothing about the biology. The rule excludes its own BUB1 positive control at +0.192 and had power 0.514 against its own pre-registered minimum effect
`PRE-REG-D`, pathogenicity placement of the missense allele	A model placing it outside the range spanned by two characterised low-abundance alleles contradicts the placement	Two of three calibrated predictors contradict	The assumption that the allele is pathogenic in the same way as the published class. Superseded as the gate by R2-A, which asks the abundance question with an abundance instrument (Section 3)
R2-A gate strength	Descriptive check, declared in advance	On BUBR1's own 8 published alleles the method concords 5 of 8, AUC 0.60 (95% CI 0.08 to 1.00), permutation p = 0.39	That the structural prediction can be cited as evidence. It is a calibrated prediction with weak support
R2-B, niacin deficiency hypothesis	NAD+ rise expected to be confined to deficient subjects	"Blood NAD+ increased in all subjects, up to 8-fold" (verbatim, verified); non-deficient controls also rose, by an unverified amount	The claim that the NAD+ rise requires a deficient baseline
R2-B, NMN dose-gap hypothesis	Mouse in vivo dose expected to be unreachable in humans	Human-equivalent dose 40.5 mg/kg/day; R = 0.10 to 0.51 against human trial exposures	The claim that the in vivo NMN arm is unreachable on dose alone. Route mismatch remains

Test	Pre-registered rule	Result	What it rules out
R2-C, class I HDAC route	F1: a source showing BubR1 stabilisation plus a working checkpoint on class I HDAC inhibition	F1 not met across the 14 records retrieved at the frozen run on 2026-10-20; 6 primary sources point the other way. The same query re-run returns 25 unfielded and 11 fielded, not yet re-adjudicated (Section 8.2)	The HDAC2/3 route as a mechanism, and 4-PBA's HDAC activity as a second mechanism in its favour

Table 5. Clinical-trial context (`T2-TRIAL-01`). ClinicalTrials.gov API v2, query "nicotinamide mononucleotide OR NMN OR nicotinamide riboside", statuses recruiting, active, completed and not yet recruiting: 148 studies. Examples as registered: NCT04691986 (NR, sarcopenia), NCT03579693 (NR with CoQ10, CKD with sarcopenia and frailty), NCT06208527 (NR, frailty), NCT07693543 (NR, CKD), NCT07144527 (NMN, exercise tolerance and frailty). **No registered trial in *BUB1B*, SIRT2, MVA or aneuploidy.** This supports the feasibility of a muscle or kidney endpoint design; it is not evidence of benefit in this disease.

9.3 Criterion C4 fails its own validity test: NON-DISCRIMINATING

Pre-registered as a compartment-validity test of our own instrument, with the firing rules fixed before the sources were opened. **Outcome: C4 is declared NON-DISCRIMINATING**, fired by rule D2 in the headline compartment. Three further flags fired independently: D1 in mouse liver, NOT-TRANSFERABLE-WITHOUT-CORRECTION in every compartment measured, and ABOVE-WINDOW on the dose-consistency check. These are four separate defects, not one restated.

What was measured. Intracellular nicotinamide in human blood cells (PBMC) is 2.6 to 7.1 uM across all 11 observations (VERIFIED, Trammell 2016, PMID 27721479, read at PMC5062546, molar units in the source). Mouse liver is about 230 uM (VERIFIED, conversion stated in the same paper). Human skeletal muscle is recorded as 86.5 pmol/mg and left UNCONVERTED, because the measuring paper states no conversion of its own and the pre-registration forbids silent cross-paper conversion. The compartments disagree by a factor of about 30 to 45, and C4 used none of them: it used plasma.

Why the rule fired. In the headline compartment, oral NR at 1,000 mg produced **no increase** in intracellular nicotinamide (VERIFIED, two independent statements in Trammell 2016: "Whereas PBMC Nam was never elevated by NR", and no dose-dependent increase across 100, 300 and 1,000 mg). Independently replicated in a different compartment and cohort (VERIFIED, Elhassan 2019, PMID 31412242: muscle nicotinamide 92.0 versus 86.5 pmol/mg, p = 0.96; blood 10.60 versus 9.50 uM, p = 0.41). Largest increment in the headline compartment is 0 against a baseline of about 5 uM, which meets the D2 condition.

Why the PASS readings are unreadable. The C4-PASS figures are plasma (NMN 0.135 uM, NR 0.358 uM). The measured baseline **inside** human blood cells is 2.6 to 7.1 uM, that is 7 to 50 times larger, and neither supplement raised it. A ratio computed from the plasma number therefore describes a term that is not the dominant one in the compartment where SIRT2 works. The criterion would return a PASS for these molecules whatever the intracellular concentration did.

Transferability. The 70 +/- 5 uM IC50, reported in the paper's text as SIRT2 while its own Table 1 column order is internally inconsistent between its nicotinamide and EX527 rows, so that the prose assignment to SIRT2 is the only self-consistent reading (verified in the free full text at PMC6901289), was measured at 1 mM assay NAD+. Measured NAD+ is about 12 to 18 uM in human PBMC and 20.9 to 47.8 uM in human blood, that is 21 to 83-fold lower, so the flag fires in every human compartment. This is a magnitude-only flag: no primary source on the NAD+ dependence of nicotinamide inhibition was read, so it does **not** claim the true inhibitory concentration is higher or lower in cells. It claims that a 21 to 83-fold mismatch in the cofactor the mechanism runs on was never checked before the IC50 was used as a threshold.

Dose window. Pre-fixed window 500 mg/day to 3 g/day, fixed before the source was opened. The primary

comparator located is the ONTRAC phase 3 trial, 386 participants on "500 mg of nicotinamide twice daily or placebo for 12 months", that is 1 g/day (PMID 26488693, verified at abstract). The exposure behind the nicotinamide exclusion came from a 6 g oral dose given as a radiosensitiser, which is **2 times the upper edge of the window and 6 times the best-evidenced trial dose: ABOVE-WINDOW**.

What this changes, and what it does not.

1. **Nothing is removed and nothing is reinstated.** The recorded results stay in the register exactly as written. No amendment is proposed.
2. **The nicotinamide exclusion stays recorded, and it is not evidence of harm.** It is a mechanism-direction reading on an instrument that cannot discriminate, taken at an exposure above the human dose window. What is not established is that a 6 g oral dose leaves intracellular nicotinamide unmoved; at 1.28 mM plasma, 180 to 490 times the measured intracellular baseline, a rise is the expected outcome. The arithmetic of the exclusion is not refuted here. The instrument's ability to discriminate is.
3. **The NMN and NR pass is not evidence of safety** on this axis, or on any axis.
4. **The SEPARATE decision for niacin and nicotinamide is unaffected**, having been independently carried by criteria C1, C2 and C3, each sufficient on its own under the frozen rule.
5. **Niacin's C4-UNRESOLVED reading is reinforced**, not resolved: this test shows the criterion could not have resolved it.
6. The pre-registration of `T2-C4-PRECURSORS` recorded this weakness in advance, in its own section 3.2, and required that any such argument apply to **all** C4 readings including the nicotinamide exclusion rather than being used selectively. This result is that argument, measured, and applied symmetrically as that clause demands.

9.4 Power of the DepMap correlation tests: one null stands, one is relabelled

Pre-registered, with the informativeness rule fixed in advance. **Outcome: INFORMATIVE-AT-T_A-ONLY.** The run reproduces every published value exactly from the public file (SIRT2 +0.0432 p 0.138; BUB1 +0.1924 p 2.8e-11; TRIP13 +0.0978 p 7.8e-04; genome 95th percentile +0.0885, median +0.0034; n = 1,178 lines, 17,916 genes).

`T2-DEPMAP-01` is NOT relabelled. Its threshold is a positive rho at $p < 0.05$, and by simulation (2,000 bootstrap resamples, injection into the real SIRT2 vector) the test had **power 1.000** to detect rho 0.20 at that threshold and 0.934 to detect rho 0.10. Five of five pre-named positive controls reach it (MAD2L1 +0.204, CDC20 +0.118, TTK +0.110, TRIP13 +0.098, BUB3 +0.094). BUB1 (+0.192) also reaches it and is **not** one of the five: the same pre-registration declares it a sixth reference point that is not counted as a control, and the result file records it separately. SIRT2 is the only gene in that set that does not reach the threshold. The objection that the test could not have detected a real relation is refuted by direct simulation and by the controls. A robustness injection into the NAMPT vector instead of SIRT2 lands within 0.01 at every level, so the power is a property of the BUB1B vector and the sample size, not of SIRT2's flatness. **"Falsified at the pre-stated threshold" stands as written.**

The limit on that. Power to detect an effect the size of the one actually observed (0.043) is only **0.341**. The test cannot exclude a true relation of about that size. It excludes a relation of the size the project itself called interesting. The scoped claim, "SIRT2 is not a fitness-level partner of BUB1B in this dataset", is what the result supports.

`PRE-REG-B` IS relabelled. Its rule (rho >= 0.20 and genome percentile >= 95) had power **0.514** against its own pre-registered minimum effect. The genome-wide maximum against BUB1B is +0.257 (CCNB1), so the floor sits within 0.06 of the largest value the data can produce, and **the rule excludes its own BUB1 positive control at +0.192**. Four of the five real checkpoint partners fail it too. Its finding that no candidate is co-dependent is therefore **a statement about the rule, not about the biology**, which is the register's own note at the time ("strict by construction"), now quantified.

Neither finding revives the SIRT2-to-BubR1 hypothesis. A test that could not answer is not evidence

for a hypothesis, and neither DepMap test addressed the acetylation mechanism at all. The pharmacological support in Section 4.5 is a separate claim on a separate instrument.

Power guide for a hypothetical human study, illustrative only (`T2-POWER-01`). For a two-arm parallel design, two-sided alpha 0.05, about 64 subjects per arm give 80% power at a standardised effect of d = 0.5 (176 at d = 0.3; 26 at d = 0.8). No effect size or standard deviation was taken from any published source, so this is a guide and not a sample-size estimate. With a single child, no realistic arm size reaches adequate power for a small to moderate effect.

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10. Tests and falsifiers

Test ID	Mechanism arm	Readout	Falsified if
`T2-EXP-GATE`	**The gate (Section 3), prerequisite for all of the below**	Steady-state BUBR1 level, cycloheximide-chase half-life, and **rescue at matched expression** scored on chromosome alignment and mitotic index, for the missense allele alongside wild type, I909T, L1012P and Q921H as an in-domain wild-type-level control; dose-response arm able to detect a non-monotonic response	Wild-type-comparable level **and** wild-type-comparable half-life (the allele is not abundance-limited); **or** rescue at matched expression fails to restore alignment (the allele is function-defective, and every arm below is void); **or** alignment or PCS worsens with rising expression (dominant-negative: the worse-with-more outcome)
`T2-EXP-PROTEO`	Chemical chaperone (4-PBA)	Steady-state level and half-life as separate readouts, with a proteasome arm, a stage marker or synchronisation, and a PCS or interphase-FISH arm	No change in steady-state level and none in half-life; or any rise in PCS or missegregation rate
`T2-EXP-HSP`	Heat-shock co-induction (arimoclomol)	HSP70 induction as the engagement control, plus BUBR1 level and half-life, in paired stressed and unstressed arms	No HSP70 induction at a reached exposure; or induction with no BUBR1 change; or a rise in PCS rate
`T2-EXP-NAD`	NAD+ raising (niacin, NMN, NR)	NAPRT and NMRK1/2 expression in the target cell type as a first-step go/no-go; then BUBR1 level and half-life at a **measured intracellular** NAD+ rise, with intracellular nicotinamide in the same cells and SIRT2 inhibition as the mechanism control	No intracellular NAD+ rise; or a rise with no BUBR1 change; or an intracellular nicotinamide rise into the inhibitory range; or a rise in PCS rate

These four IDs replace the five used in earlier drafts, none of which corresponded to a register entry. `T2-EXP-PP2A` is retired with the exclusion of the PP2A mechanism.

Four design corrections, made because the previous set did not discriminate.

1. **The same cycloheximide chase was the readout for five arms**, so it tested a shared premise and collapsed the arms rather than separating them. It is now the **gate's** readout, where testing a shared premise is the point, and each arm carries its own engagement control (HSP70 induction, intracellular NAD+, proteasome dependence).
2. **A chase measures degradation only.** A mechanism acting on synthesis would show no half-life change while working, so "no change in half-life" alone is no longer a falsifier: steady-state level and half-life are separate readouts with separate falsifiers.
3. **Cycloheximide is confounded for a heat-shock co-inducer** and BubR1 turnover is strongly cell-cycle dependent, so a stage marker or synchronisation is required and asynchronous fibroblasts alone

are not acceptable.

4. **A one-sided falsifier in the wrong direction has been removed.** The retired PP2A readout used Aurora B inhibition as a positive control, which lowers mitotic index in the same direction a PP2A-B56 activator pushes, so the predicted harm would have registered as support.

Effect size: the previous proposal was too small to mean anything, and it is withdrawn as a threshold. Section 4.2 anchors aneuploidy below roughly 50% residual BUBR1 with PCS present at every level tested; the published low-abundance class sits near 10 to 20%; Section 4.3's rescue was to wild-type-comparable levels. On that anchor a 1.25-fold rise moves roughly 20% to 25% and crosses nothing, and 1.25-fold at three replicates by Western densitometry is out of statistical reach. The go/no-go numbers in the previous revision (ratio at least 1.25 over vehicle with a 95% CI lower bound above 1.0; viability loss 20% or less; controls moving in the expected direction) appear in no register entry and no frozen plan. They are withdrawn as thresholds and recorded as an open requirement: a target effect size must be derived from the restoration demonstrated in PMID 20516114 Fig 4D, and the replicate count from a stated assay coefficient of variation, and both must be frozen before any experiment runs if they are to count as pre-registered.

10.1 The measurement that tests this report's own premise

Section 5 states the genotype class as one truncating and one missense *BUB1B* allele, and states that the trans configuration is **inferred, not measured**: read-backed phasing returned UNRESOLVED because no sequencing template in the data spans the gap between the two sites. Everything in this report stands on that inference. If the two alleles sat on the same chromosome, one *BUB1B* copy would be intact, and whether residual BUBR1 is the limiting quantity would be a different question with a different answer.

A long-read run on existing material settles it. One molecule covering both sites shows the two alleles on the same strand or on opposite ones, which decides the configuration directly, with no parental sample and no cell experiment. It is the cheapest decisive measurement on this report's list, and the gate in Section 3 is worth running only if the inference holds.

The same run returns four things that short fragments cannot give. Each is a limit of the data rather than of the analysis, which is why no reanalysis of the existing reads recovers them:

- **Structural variants.** In short-read data the evidence for a breakpoint or a large insertion is the absence or the displacement of reads, not a read that crosses the event. A long read crosses it.
- **Insertions and deletions.** Short-read calls are anchored by position, so the anchor and the consequence can disagree. A read that contains the whole allele removes the convention.
- **Paralogous and duplicated regions.** A short fragment from a duplicated region cannot be placed uniquely. A read that reaches the unique flank can.
- **Base modification.** A nanopore run reports it from the same pass, with no separate assay and no further sample. No role for base modification in this condition is established here; it is listed because it costs nothing to collect alongside the rest, not because this report predicts a finding.

What it does not settle. Phase is not function. Knowing the two alleles are in trans does not say whether the missense allele is abundance-limited, which is what Section 3 asks and what the gate measures, and it does not measure what the substitution does to the protein. A long-read run removes an inference from under this report. It removes no experiment from the table above.

The full laboratory protocol, with cell system options, conditions, controls and blinded scoring, is held separately and contains no results.

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11. Safety

Nothing in this section is a safety clearance for anything, and nothing in it is advice.

11.1 What was measured, and what it cannot say. On-target knockout dependency in 1,178 DepMap 24Q4 cancer cell lines, for 24 of 26 pre-registered candidate-target genes, with blood-lineage medians reported because marrow is an affected compartment in this phenotype. A knockout dependency is a loss-of-function proxy; for substrate-feeding arms (niacin, NMN, NR), for a reported activator and for a reported co-inducer the knockout direction is **opposite** to the molecule's direction. These are transformed cancer lines, never normal marrow. **No flag is a safety claim about any molecule or any person, and no absent flag is a safety clearance.** The flags removed nothing and now order nothing.

11.2 An adverse human blood-lineage signal in our own cited source. PMID 32386566 is the source this report uses for niacin's verified human NAD+ rise, and it was read in full. It also reports, verbatim: **"Some patients showed anemia tendency, while muscle strength and mitochondrial biogenesis increased in all subjects."** The previous revision cited this paper twice, built an entire blood-lineage DepMap analysis around marrow suppression as the main concern for this phenotype, and did not carry that observation across. It is carried here: a human observation of an anemia tendency, in the compartment this report names as the one to watch, attached to the NAD-raising mechanism family. It sits alongside the measured dependency signals that motivated the concern (NAMPT fraction 0.296 with blood median -0.936; NMNAT1 blood median -0.911). Normal marrow remains untested and is the next required safety test for any NAD-raising mechanism.

11.3 Label warnings, replacing the absence of a boxed warning. The previous revision used "no boxed warning" as a reassurance proxy. That is retired: the absence of a boxed warning is not a safety finding, and the labels this report calls the reference for known drug safety were not reproduced. Verified label text, openFDA, queried by brand name on 2026-10-20:

Molecule	Label warnings as written	Paediatric label
Niacin, immediate-release prescription (NIACOR)	**Liver dysfunction:** "Cases of severe hepatic toxicity, including fulminant hepatic necrosis have occurred in patients who have substituted sustained-release (modified-release, timed-release) nicotinic acid products for immediate-release (crystalline) nicotinic acid at equivalent doses", with scheduled transaminase monitoring, and active liver disease or unexplained transaminase elevation as contraindications. **Skeletal muscle:** rare rhabdomyolysis with concomitant HMG-CoA reductase inhibitors at lipid-altering doses. **Adverse reactions include** atrial fibrillation and other arrhythmias, orthostasis, hypotension, jaundice and abnormal liver function tests, mild to severe cutaneous flushing, **decreased glucose tolerance, hyperuricaemia and gout**, toxic amblyopia and cystoid macular oedema	Use not established at 16 years and under

Molecule	Label warnings as written	Paediatric label
Arimoclomol (MIPLYFFA)	**Hypersensitivity reactions:** urticaria and angioedema reported, within the first two months; discontinue on severe reactions. **Embryofetal toxicity:** "May cause fetal harm", with post-implantation loss and structural abnormalities in rats and rabbits. **Increased creatinine without affecting glomerular function:** mean rises of 10 to 20%, so creatinine-based renal assessment is unreliable. **Juvenile animal toxicity:** increased renal pelvic dilatation "at all dose levels" in juvenile rats dosed from postnatal day 7	Established at 2 years and older; not established below 2 years
Sodium phenylbutyrate (BUPHENYL label text)	**Sodium load:** 62 mg sodium per tablet and 125 mg sodium per gram of sodium phenylbutyrate; "should be used with great care, if at all, in patients with congestive heart failure or severe renal insufficiency, and in clinical states in which there is sodium retention with edema". Caution in hepatic or renal insufficiency and in inborn errors of beta oxidation. **Adverse reactions:** amenorrhoea or menstrual dysfunction in 23% of menstruating patients; decreased appetite; body odour and taste aversion; **haematologic: aplastic anaemia and ecchymoses each reported in one patient** of 206; neurotoxicity reported with intravenous phenylacetate in cancer patients	Adult and paediatric (urea cycle disorders); tablets not recommended for neonates, infants and children up to 20 kg

Two of the three carry a marrow-relevant item (the niacin anemia tendency in Section 11.2; a single reported case of aplastic anaemia on the phenylbutyrate label), and both are reported rather than weighed. Among the **excluded** class I HDAC inhibitors, the labelled myelosuppression is explicit and is one reason the exclusion is not a loss: romidepsin and belinostat labels state thrombocytopenia, leukopenia and anaemia, and the one member with a chronic paediatric indication states that it "can cause dose-related thrombocytopenia and other signs of myelosuppression, including anemia and neutropenia".

11.4 Cancer risk has not been assessed, and this is a cancer-predisposition syndrome. MVA carries childhood cancer predisposition (Section 2; PMID 15475955). **No mechanism or molecule in this report has been assessed for its effect on that risk, by us or by any source we found.** This is the most consequential gap in the document and it is named as a required test, not as a residual caveat. The specific questions, each unanswered: whether raising NAD+ supplies a substrate advantage to a transformed clone in a child already predisposed to embryonal tumours and leukaemia; whether chronic HSP70 induction acts as a pro-survival chaperone in such a clone; whether a chemical chaperone with inseparable HDAC activity alters missegregation rate in cells that already missegregate, which is the direct mechanism by which an intervention here could **increase** tumour risk (PMID 16857700 shows HDAC inhibition is an aneugen in human cells). The required test is a PCS or micronucleus and aneuploidy-rate readout in the same experiment as any abundance readout, which is why every falsifier in Section 10 now carries a PCS arm. The word immunosuppression appeared nowhere in the previous revision's safety section; the one arm that carried chronic lymphodepletion has since been excluded on mechanism (Section 8.1), but the general point stands that tumour surveillance is a safety axis this report has not addressed.

11.5 No stop rule is claimed that does not exist. The dependency flag never removed anything. The

only exclusions in this report come from mechanism direction (PP2A, class I HDAC, nicotinamide's C4 reading), from T2-P4 target-class dependency, from regulatory class, and from scope. An earlier draft stated a removal rule at "dependency fraction above 0.5"; no such rule exists in the register and it is withdrawn.

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12. Pre-registration ledger

Every number in Sections 3 to 11 traces to one of these entries. "Rule fixed before data" means the decision rule was written and hashed before the inputs were opened. Hashes are 12-character sha256 prefixes; full values are in HYPOTHESIS_REGISTER.md.

Test ID	Question	Rule fixed before data	Result	What it rules out	Files (sha256 prefix)
PRE-REG-R2A	Is the missense allele in the published low-abundance class?	ThermoMPNN ddG on AlphaFold models, calibrated on PTEN and TPMT VAMP-seq; gates and thresholds fixed; frozen 11:23:17Z, first download 11:23:36Z	ABUNDANCE-LIKE under the rule, applied mechanically; evidence weak (calibrated probability 0.53, margin 0.36 kcal/mol below method RMSE)	Nothing. It is the gate, and it does not establish the abundance premise	pre-reg be9390e6eafa; result 62a3405d3bd5
VERIFY-R2A	Does R2-A survive an adversarial audit?	Independent recomputation through a different code path; leakage scan; sensitivity analysis	CONFIRMED-WITH-CAVEATS. All 58,760 ddG values reproduce at max difference 0.000000; gate would have failed on 7 of 10 possible control pairs; recommended wording adopted verbatim in Section 3	Citing R2-A as support rather than as a weak prediction	verification 202d669b794c
T2-EXPOSURE-01 (R2-B)	Can the proposed mechanism be engaged at the exposure the literature establishes, applied symmetrically to all arms?	Frozen 11:18:24Z before any figure was opened; ratio limbs, direction node rule and falsifiers fixed; less favourable limb scored	PP2A arm E-FAIL on both limbs and EXCLUDED; all six other arms E-UNRESOLVED; 5 of 7 arms have no establishable C_demo	The PP2A mechanism; and any ordering of the surviving arms by engagement evidence	pre-reg 338b0d6078d9; result a9449002b434
T2-HDAC-R2C	Does the HDAC2/3 route admit any approved class I HDAC inhibitor?	Frozen 11:17:46Z; universe, criteria C1 to C6 and falsifiers F1 to F5 fixed in advance	**Zero entrants.** Class excluded on direction on six primary sources; F1 not met; HDAC3 the only flagged isoform of 11	The HDAC2/3 route as a mechanism; and 4-PBA's HDAC activity as a point in its favour	pre-reg e97a150047ca; result 7494a26b4382

Test ID	Question	Rule fixed before data	Result	What it rules out	Files (sha256 prefix)
PRE-REG-NONCHOICES	Which proposed mechanism classes does the published record already exclude?	Frozen 2026-10-20T11:42:00Z before the screen ran; two arms, gene set, inclusion rule, verdict definitions and the DepMap flag rule all fixed in advance	25 verdict rows from 24 numbered classes: 11 EXCLUDED, 6 NOT TESTED, 8 OPEN, of which 6 bear directly on raising BUBR1. 226 rows, of which 88 (39%) reach the universe through the literature arm alone with no assigned mechanism, and 114 carry no assigned mechanism once the 26 also-considered rows are included. 60 targets measured, controls reproduce exactly	Ten mechanism classes beyond those already excluded; the chemical-chaperone rationale for one of our own arms (zero records); the claim that the heat-shock class has no BubR1-specific rationale; and any reading of curcumin or vitamin C as a live option	pre-reg ed19c9360d30; section a3917662993a; evidence 3595d87d4a4f; batches 61c1fca70c2f
PRE-REG-R2D	Does criterion C4 measure the quantity it reasons about?	Compartment priorities, firing rules D1 and D2, transferability and dose-window checks all fixed before sources were opened	**C4 NON-DISCRIMINATING**; three further flags fired independently; nicotinamide's exposure ABOVE-WINDOW	Reading the nicotinamide exclusion as evidence of harm, or the NMN and NR pass as evidence of safety	pre-reg a37962431d2c; result e797cd4e6812
PRE-REG-R2E	Could the DepMap correlation tests have detected their own pre-registered effects?	Thresholds T_A and T_B, injection method, control set and informativeness rule I1 and I2 fixed in advance	INFORMATIVE-AT-T_A-ONLY. T2-DEPMAP-01 power 1.000 at T_A, 5 of 5 controls pass, not relabelled; PRE-REG-B power 0.514, relabelled	For T2-DEPMAP-01: the objection that it was underpowered at its own threshold. For PRE-REG-B: its own negative as a biological statement	pre-reg a37962431d2c; result e797cd4e6812
T2-P4	On-target dependency of the first candidate-target set	Flag if fraction with Chronos < -0.5 is >= 0.25; descriptive only; no flag is a safety claim	7 of 11 genes flagged (MTOR, CDC20, PLK4, ANAPC4, UBE2C, KIF18A, NAMPT)	mTOR, APC/C, PLK4 and KIF18A as candidate target classes	HYPOTHESIS_REGISTER.md
T2-P5	On-target dependency of the targets T2-P4 did not cover	Same flag rule, unchanged; absent genes reported as "not measured", never as 0	24 of 26 genes measured; PPP2CA, PPP2R1A, HDAC3 flagged	The claim that these arms have no measurable target; the claim that niacin and nicotinamide share a safety placement	HYPOTHESIS_REGISTER.md
T2-DEPMAP-01	Does SIRT2 knockout fitness co-vary with BUB1B?	Positive rho predicted; falsified if rho is near zero	rho +0.043, p = 0.14; stands after R2-E	A shared fitness-level stability route in cancer lines	HYPOTHESIS_REGISTER.md

Test ID	Question	Rule fixed before data	Result	What it rules out	Files (sha256 prefix)
PRE-REG-B	Any co-dependent lever gene for a combination?	rho >= 0.20 and genome percentile >= 95	None of 17,916 genes qualifies; **relabelled** by R2-E as a property of the rule	Nothing about the biology	HYPOTHESIS_ REGISTER.md
PRE-REG-D	Pathogenicity placement of the missense allele	Descriptive placement only; a model placing it outside the two-allele range contradicts the placement	2 of 3 calibrated predictors contradict	The assumption that the allele is pathogenic in the published manner. Superseded as the gate by R2-A	HYPOTHESIS_ REGISTER.md
RANKING-RULE-T2	How were candidates ordered?	Grade first; ties by fewer flagged targets; a supplement sits above a prescription drug only on a strictly higher grade	Applied, corrected (APPLICATION-FIX-T2-01), and its ordinal output now **withdrawn** (Appendix A)	Ordering by preference rather than by rule; and, now, ordering at all	HYPOTHESIS_ REGISTER.md; audit 5dac563cfb1e; order derivation ea70c0fa0cc8
APPLICATION-FIX-T2-01	Was the rule applied correctly?	Rule text unchanged; grades unchanged; correction to the application only	Order corrected, then the ordinal output withdrawn entirely	The earlier order, and the invented clause "no supplement is above an OTC drug"	HYPOTHESIS_ REGISTER.md
T2-CLASS-01	Are niacin and nicotinamide one candidate, and in which class?	Five criteria (C1 regulatory, C2 entry enzyme, C3 safety placement, C4 mechanism direction, C5 direct evidence); any one difference gives SEPARATE; C4 "opposes" drops a molecule	SEPARATE, then DROP nicotinamide on C4; niacin is prescription and is not dropped. SEPARATE unaffected by R2-D; the C4 half is now non-discriminating	Treating the two molecules as one candidate	HYPOTHESIS_ REGISTER.md
T2-GRADE-01	Is PMID 25285631 a "published contradiction" under the ranking rule?	E, M, D and I all MET required; S a weakening; INDETERMI-NATE gives counter-evidence, not contradiction	NMN G3 under the definition; verification judged it **G3-contested** and found criterion D structurally favours promotion	A clean G3 for NMN. With the ordinal list withdrawn, the grade no longer orders anything	pre-reg e18655b31d4e; result 48a9b709b7ce; verification b6ad7006950d

Test ID	Question	Rule fixed before data	Result	What it rules out	Files (sha256 prefix)
`T2-C4-PRECURSORS`	Does criterion C4 drop NMN or NR?	IC50 verified at source; pass bound 0.1 x IC50; conflicts and unresolved cases pre-specified	C4-PASS both (ratios 0.0019 and 0.0051 against 70 +/- 5 uM, a value the source states as SIRT2 in its text, its Table 1 column order being internally inconsistent between two rows so that the prose assignment is the only self-consistent reading, verified at PMC6901289). **Not evidence of safety**, per R2-D	A mechanism-direction objection at the plasma figures used; nothing more	pre-reg `66aad129e448`; result `87ddeef92d7e`
`T2-C4-NIACIN`	Does the same frozen criterion drop niacin?	Addendum reuses the frozen rule verbatim; frozen 2026-10-20T09:37:12Z, before any niacin PK figure was searched	**C4-UNRESOLVED**; niacin not dropped. NAD+ rise verified (PMID 32386566); no verified plasma nicotinamide figure after oral niacin exists in any reachable source	Nothing. It records an open direction question, now reinforced by R2-D	addendum `5c461d4f817d`; result `8543cf9cafdb`
`T2-TRIAL-01`	Is a trial design feasible from the registry?	Query string and status filter recorded	148 studies; none in *BUB1B*, SIRT2, MVA or aneuploidy	A claim of existing human evidence in this disease	HYPOTHESIS_REGISTER.md
`T2-POWER-01`	What arm size would a trial need?	Illustrative guide only; no effect size taken from any source	64 per arm at d = 0.5 for 80% power	Using these numbers as a sample-size estimate	HYPOTHESIS_REGISTER.md

Test ID	Question	Rule fixed before data	Result	What it rules out	Files (sha256 prefix)
Citation audit	Does every cited source support the sentence it is attached to?	Two independent checks, the second re-deriving from primary sources before opening the first	15 support exactly, 6 partially, 3 pending markers resolved, and PMID 25672491 mischaracterisation found and corrected in Section 4.5. **One DOES NOT SUPPORT row falls on this report** (the read-through exclusion, withdrawn in Section 8.3); a second DOES NOT SUPPORT row is graded against the Track 1 methods draft, not this report. The check's own counts table records 1 while its row-level grades show 2, a discrepancy in the source that we flag rather than resolve	Three unresolved pending markers, the "biallelic loss" wording, and the single-laboratory claim	`4d3a33d2f1fb`; second check `bd91b0a09d00`

12.1 Disclosures about the pre-registration itself

Stating these is the point of keeping a register.

1. **`T2-DEPMAP-01`** states its hypothesis inside the result entry rather than in a separate pre-registration written beforehand.
2. **`T2-TRIAL-01`** has no pre-registration entry at all; its query string and status filter are recorded with the result.
3. **`RANKING-RULE-T2`** was written after the `T2-P4` results that its safety tie-break consumed, and was not amended afterwards. Its ordinal output is now withdrawn (Appendix A.1), so the tie-break that consumed those numbers no longer produces anything.
4. **T2-GRADE-01's criterion D** bundles direction with auditability, so a closed-access paper can never demote a claim under it. This was not disclosed in that pre-registration and was found by verification.
5. **PRE-REG-R2A's freeze log is self-attested on one machine.** The mtime chain and hash agreement are consistent with the claimed order but there is no external notarisation.
6. **PRE-REG-R2A's F4 allele table and its "5 of 8 concordant, AUC 0.60" metric were built after the result.** The pre-registration promised only a descriptive table, so this is compliant, but the metric is post-hoc and must be read as such. Two sub-criteria in the decision script were hardcoded after manual inspection rather than computed; one was re-derived independently and verified, the other cannot change the outcome.
7. **`PRE-REG-R2C` scored three molecules as reaching their target on a label mechanism-of-action statement** rather than on a computed free concentration; panobinostat's marketing status is inferred from two 404 responses, not from a withdrawal notice.
8. **The go/no-go thresholds** in the previous revision are withdrawn as thresholds (Section 10) and nothing has replaced them yet.
9. The 1.2 uM SIRT2 IC50 figure that could move both precursors to C4-AMBIGUOUS is **unverified** and comes from a closed-access record. After R2-D this matters less, since the criterion is non-discriminating either way.

10. **One figure used in the previous revision was the wrong analyte.** The 0.135 uM and 0.358 uM "NMN and NR" exposures are plasma **nicotinamide**, not plasma NMN or NR. Plasma NMN after oral dosing is not established in any source we reached.
11. **One figure is a unit conversion rather than a script output.** The 1.28 mM nicotinamide plasma figure is the source's 156.4 ug/mL converted at the molecular weight, and that conversion is recorded in the pre-registration log rather than produced by a script. The front matter's and Section 17's statement that every computed number comes from a script therefore has this exception, and it is listed here because that statement points at this section.

12.2 Reproducibility

- Pre-registration log, also called the hypothesis register: HYPOTHESIS_REGISTER.md, append-only, with input checksums and decision rules fixed before each run. Identical copies held on two machines, hashes compared at each sync. The hash current at the time this revision was written is recorded in the register's own final entry rather than quoted here, because quoting it inside a file that the same append then changes cannot be kept true.
- Code: the GitHub repository (public after the hackathon closes), with the DepMap, lever-screen, dependency, HDAC-screen, power-simulation and structural-prediction scripts.
- **Repository:** the Markdown sources of both this report and the Track 1 methods report are in the repository alongside the analysis code. The repository contains no sequencing data of any kind: no VCF, FASTQ or BAM file, no per-variant or genotype-scale table of the child's calls, and no frequency or score table of the child's variants. The hypothesis register is not in the repository.
- Data: DepMap 24Q4 (public, CRISPRGeneEffect.csv and Model.csv), ClinicalTrials.gov API v2, Drugs@FDA and drug labels via openFDA, DailyMed, EMA public pages, PubMed, PMC and Europe PMC, Human Protein Atlas, PubChem, UniProt, InterPro, AlphaFold DB, MaveDB.
- Only gene, protein, drug and mechanism terms, PMIDs, DOIs and public accessions were sent to any external service. **No patient-derived data left the local machines.** The AI disclosure above records that local sessions saw child-derived data; that is consistent with this statement and the previous revision's wording was not.

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13. Limitations

- All BUBR1 mechanism evidence here is cell-based, mouse-based or structural prediction. No human BUBR1-rescue data exist in our sources, and no molecule named here has any patient-derived or human BUBR1 data.
- The gate is a prediction, not a measurement, and its evidence is weak by its own verifier's assessment.
- DepMap lines are cancer lines. Blood-lineage results are not normal-marrow results, and two arimoclomol target genes were not measured at all.
- Five of seven mechanism arms have no establishable concentration at which the mechanism was demonstrated on BUBR1, so no engagement ratio exists for them. Free plasma concentration, where it was computed, is a proxy for concentration at the kinetochore and was not measured there by any source.
- One of our own instruments (criterion C4) is declared non-discriminating by its own validity test.
- **Six papers are closed access on every route checked**, and a later audit found that four more we had recorded as unreachable are in fact indexed as open. Genuinely closed: the four human pharmacokinetic papers that would resolve the niacin direction question (PMID 17725178, PMID 17463214, PMID 20562093, PMID 27567458), the paper that decides the NMN grade (PMID 25285631), and the single source behind the SIRT1 transcriptional route (PMID 35639235, which is also where transcript against protein would be settled). Recorded as unreachable during the run, but with an open copy indexed and to be re-read before the next revision: PMID 17717597 (free at the publisher and in PMC1950458), PMID 10677536 (in PMC26515), the second IC50 record PMID 26696402 (repository copy) and the curcumin selectivity paper PMID 15383533 (publisher open access, although the route we tried returns HTTP

403). No verdict turns on that correction: this report's definition of a demonstration concentration requires it to be measured on BUBR1, and neither PMID 17717597 nor PMID 10677536 measures BUBR1 at any access status, so both arms stay E-UNRESOLVED on the evidence rather than on the access.
- The HDAC exclusion rests on cancer-line pharmacology at fully inhibitory concentrations, none of it in *BUB1B*-mutant cells.
- **The mechanism-class screen is bounded by its own search.** Its database arm effectively rests on one source, because a pre-registered target-type filter excluded protein-complex and protein-family targets and returned a single approved drug from the other database. **Eighty-eight of 226 molecules (39%) reach its universe through the literature arm alone with no assigned mechanism**, because the published literature pairing *BUB1B* with drugs is mostly about the checkpoint predicting cytotoxic sensitivity rather than about raising BUBR1, and the frozen queries cannot separate the two kinds of paper. A further large share of the database rows rest on a claimed target that no primary source supports, with empty interaction-type records and confidence scores as low as 0.003. It is not a systematic review: no PRISMA flow, no second screener, no inter-rater agreement. Any mechanism whose rationale has never been written down is invisible to it (Section 15.2).
- **A coverage gap that falls on two of our own arms.** One slice of about 45 proteostasis and chaperone molecules in that screen was never evaluated drug by drug, and the three classes it covers are HSP90 inhibition, chaperone co-induction and chemical chaperones. Two of those three are the arms carried forward in Sections 7.2 and 7.3, so those arms carry a coverage gap in the molecule screen on top of their evidence gap.
- **Our own direction rule has a gap, and we are not patching it after the fact.** The frozen definition of AGAINST is to lower BUBR1 protein, or to silence, weaken or override the checkpoint. It has no limb for an agent that **engages** the checkpoint and still causes missegregation through a spindle geometry the checkpoint cannot sense, which is what colchicine and curcumin do. Those molecules therefore stay individually unresolved even though aneugenicity is the documented risk and aneuploidy is the disease endpoint. Reading the definition more loosely after seeing the drug is what pre-registration exists to prevent, so any fix must be a recorded amendment (Section 15.6, Appendix B class 17).
- **Our frozen exposure rule gives no tie-break between two verified demonstration concentrations**, which left one class on the boundary between two different outcomes. Both readings are reported rather than one chosen (Appendix B class 16b).
- **One exclusion in this report is procedural rather than scientific.** APC/C inhibition is excluded on our dependency rule while proteasome inhibition, which carries the same objection at the same magnitude, is not, because the frozen pre-registration forbids extending the earlier precedent to classes it did not name. The APC/C direction evidence points the other way (Section 8.3).
- Cancer risk, the central safety axis for this phenotype, has not been assessed for any mechanism here (Section 11.4).
- No patient-followable biomarker is validated for any mechanism in this disease.
- Literature marked "verified in full text" was read in full. Everything else was checked at abstract or record level and is marked accordingly.

Open items carried forward. The wet assay that decides the gate. Re-adjudication of the class I HDAC falsifier over the wider record set the same query now returns, since the printed count is retrieval-dependent (Section 8.2). Whether any chaperone effect on the checkpoint is BubR1-dependent rather than Cdc20-dependent, which is what the heat-shock arm now turns on (Section 7.3). The Track 2 share of the project's measured compute and effort, which was never apportioned per track and is reported as missing rather than estimated (Section 17, Q10). The condition under which the lagging-chromosome experiment behind the heat-shock rationale was run, which the abstract does not state and whose full text is unreachable (Section 7.3). Whether the SIRT1 promoter result is on transcript or on protein, which is the cheapest open question in the document and needs one full text plus one qPCR and immunoblot pair. Cancer-risk assessment for every arm, which no source provides. Whether the target cell type expresses NAPRT. Whether oral NMN and NR convert to nicotinamide at concentrations that invert the direction. Target effect size and replicate count, to be derived and frozen before any experiment. The six closed-access papers in Section 13, and

the four more listed there that an open copy exists for and that should be read before the next revision. A runnable reproducibility package to ship with the submission rather than a promise of a repository. Whether a PCS or micronucleus rate can serve as a patient-followable biomarker.

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14. For families and anyone reading this outside research

Plain language, and it is the most important section for anyone who is not a scientist.

Nothing in this report is a treatment, and nothing in it should change what anyone does today. This is a list of ideas about how a cell works, written so that researchers can test them in a laboratory and find out which ones are wrong.

Nothing here has been tested in cells from a person with MVA. Not by us, and as far as our searches reach, not by anyone. Every one of these ideas could turn out to do nothing. Some could turn out to make things worse, and we have written down how we would detect that.

Several of the molecules named in this report are sold over the counter as supplements, including two NAD+ precursors, and, in the wider screen in Section 15, curcumin and vitamin C. That is a fact about how they are sold. It is not a finding of this report, and for curcumin and vitamin C the report's finding is the opposite of encouraging: each was examined and neither has any published evidence of raising BUBR1, with the reasons set out in Appendix B. **Nothing in this report says that they are safe for a child, or useful for a child, or that a child should take them.** The fact that a molecule appears in this document means only that someone has published a reason to study it.

Giving a child any of these without a doctor and a research protocol could cause harm. Two separate reasons. The first is direct: these molecules have real labelled risks, including liver injury, blood-count effects, and in one case a laboratory finding showing that the same kind of drug can cause the exact chromosome problem this condition already involves. The second is less obvious and matters just as much: **if a child takes something outside a study, a future real study may become impossible to interpret for that child.** Researchers would not be able to tell what the medicine did from what the condition did. Self-administration can close the door on the answer everyone wants.

If you are a family affected by MVA and you want to help: the useful next step is not a supplement. It is contact with a clinical genetics team and with a research group that can run the laboratory test in Section 10 on patient-derived cells, because that test, not this document, is what decides whether any of these ideas is real.

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15. Non-choices: the mechanism classes the literature already excludes

This section exists because the discipline is in testing mechanisms we did not propose, not only our own. It screens every mechanism class that has been put forward as a way to raise BUBR1 protein level or restore spindle-assembly-checkpoint function, against one question: what does the published record say the mechanism actually does to BUBR1 or to the checkpoint. Per-class detail, with the deciding quotations and the experiment that would close each open class, is in Appendix B. Verdicts are MECHANISM EXCLUDED (a primary source shows the mechanism moving BUBR1 or the checkpoint the wrong way, or a frozen test of ours disposed of it), MECHANISM NOT TESTED (the deciding datum does not exist in the searched record, with the missing datum and the failed search route both named), or MECHANISM OPEN (a published rationale exists, nothing excludes it, and the deciding experiment is stated). A fourth label, NOT EVALUABLE, applies to individual molecules and means something narrower and important: **nobody proposed that molecule for that mechanism, so the claimed mechanism has no source. It never means the molecule was examined and cleared.**

15.1 How the universe was built, pre-registered before it ran. Two independent arms over a fixed 46-symbol gene set. A database arm: DGIdb GraphQL keeping approved drugs only (163 approved interactions, 124 distinct drugs), plus ChEMBL by UniProt accession restricted to human single-protein targets (33 targets, 45 mechanism records, **1 approved drug**). A literature arm: four verbatim PubMed queries, names extracted against the Drugs@FDA bulk lexicon (118 distinct names). Universe: **217 distinct molecules = 200 new + 9 already evaluated by this project + 8 excluded at assembly as non-specific chemical names.** The evaluation batches hold the 200 new plus 26 names tagged "also considered, reached outside the stated screen", giving **226 rows**. The 9 and the 8 are not in those batches, which is why 217 and 226 are not related by adding 26 to 217. The eight excluded at assembly are oxygen, sodium, calcium, potassium, amino acids, nitric oxide, cysteine and adenosine. Arm tags over the 226 rows: database arm only 93, literature arm only 88, both of those 18, the two databases together 1 (perhexiline, the single approved molecule the ChEMBL arm returned), also considered 26. Grouped into twenty-four numbered classes giving **twenty-five verdict rows**, because class 16 splits into two limbs that take different verdicts. Sixty targets were measured in DepMap under the unchanged `T2-P4` rule, all sixty returning a column and none reported as not measured, with seven re-measured controls reproducing the earlier runs exactly (Table 2b).

The drug-level result of that screen, which is the largest single piece of work in this track and returns nothing. Of the 226 rows, 181 were evaluated molecule by molecule against the frozen criteria, in five batches of 20, 14, 49, 49 and 49. **None was kept.** 29 were removed on a named criterion, 151 returned UNRESOLVED because no source demonstrates the proposed mechanism on BUBR1 at any concentration, and the one molecule provisionally kept was later closed as MECHANISM NOT TESTED (curcumin, Appendix B class 16b). The remaining 45, the proteostasis and chaperone slice, were never evaluated drug by drug and are the coverage gap disclosed in Section 13. Read with limit 2 below, that is the cleanest statement of what this screen found: a 181-molecule individual evaluation of the approved and supplement universe reachable from this gene set keeps no molecule, and UNRESOLVED removes nothing and clears nothing. Keeping no molecule is a statement about the published record and not about efficacy: it means that for none of them did a source establish the proposed mechanism on BUBR1 in the direction this report needs. The evaluation read published records and database entries and applied the frozen criteria to them; it ran no new experiment, used no cell of this child, and administered nothing. The dependency data it draws on measure what happens when a gene is disabled in cancer cell lines, which is not a test of any molecule.

15.2 Five limits, disclosed because they shape every verdict. Two of them are findings about candidate generation rather than about any molecule.

1. **The database arm effectively rests on one source.** The pre-registration fixed the ChEMBL target type to single protein before the run. ChEMBL attaches most approved-drug mechanisms to protein-complex or protein-family targets, so that arm returned exactly one approved drug. Checked as a filter property rather than assumed: the human mTOR single-protein target carries 26 mechanism records of which one molecule is at maximum phase, while the FKBP12A/mTOR complex target carries zero, so the rapalogs are reached by neither.
2. **Most of the literature arm is chemo-sensitivity co-occurrence, not a proposal to raise BUBR1.** Of 226 rows, **88 (39%) reach the universe through the literature arm alone and carry no assigned mechanism at all**: they are papers in which BubR1 or the checkpoint modulates sensitivity to a cytotoxic drug. The top names by record count are paclitaxel (35), cisplatin (31), doxorubicin (17) and docetaxel (12), and per-slice counts agree (22 of 49 rows in one slice, 24 of 49 in another, with no mechanism arm). In several source papers the therapeutic aim is the inverse of ours. PMID 34545188, verbatim: "Increased BUB1B/BUBR1 expression contributes to aberrant DNA repair activity leading to resistance to DNA-damaging agents". There the goal is to lower BubR1 to sensitise a tumour. The frozen queries cannot separate the two kinds of paper and no filter was added afterwards to make the output look better. Stated plainly: **the published literature pairing *BUB1B* with drugs is overwhelmingly about the checkpoint as a predictor of cytotoxic response, not about raising BUBR1 as a therapeutic aim.** That is the most informative thing the screen returned.

3. **A large share of the database rows rest on a claimed target that no primary source supports.** The counts are consistent across every slice that looked: 12 of 20 rows in one, 12 of 14 in another, 12 of 49 in a third, and 16 rows in a fourth of which **14 returned an empty interaction-type list on re-query, with interaction confidence scores from 0.003 to 0.236**, sourced to drug-target aggregators and repurposing hubs rather than to a pharmacology measurement. Concrete examples: bisphosphonates tagged to SIRT1 when the real target is FDPS, benzodiazepines tagged to CREBBP when the real target is the GABA-A receptor, gefitinib tagged to SIRT2 when the real target is EGFR. The field reads as "drug target" and returns association records. For those rows the frozen criteria return NOT EVALUABLE, with the meaning given above.
4. **The name-extraction implementation is looser than the frozen rule.** The rule required a name to appear as a molecule given to cells, animals or people; the implementation tests only for the name's presence in the title or abstract. The deviation can only add names, never remove one, so it cannot cause a false removal. It produced lexicon artefacts in **two distinct batches, which must not be conflated**. Eight names were caught at universe assembly and never entered the batches, and they are the reason the decomposition above subtracts 8: oxygen, sodium, calcium, potassium, amino acids, nitric oxide, cysteine, adenosine. **Four more survived into the batch tables and were identified only later, during drug-level evaluation:** ferric, ferrous, heather and nikita, strings that matched the Drugs@FDA name list inside abstracts about plant phytochemicals, bioinformatics, osteoarthritis and a case report. Two of those four resolve to real marketed products, so they cannot simply be deleted; all four sit **inside** the 226 rows and are carried as extraction artefacts rather than as evaluated molecules, while the eight sit **outside** it. All twelve names were checked against the batch tables: none of the eight is present and all four of the others are.
5. **A data-quality warning for anyone replicating this.** openFDA generic-name queries return HTTP 404 as a syntax artefact for drugs that are in fact approved. Eleven names hit it in one slice, and re-querying on brand name with two known-approved positive controls recovered both. Trusting the first pass would have produced false regulatory removals for at least five molecules. The controls caught it; anyone repeating this needs the same control.

Neither arm reaches a mechanism whose rationale has never been written down. This is not a systematic review: no PRISMA flow, no second screener, no inter-rater agreement.

15.3 The classes. Twenty-four numbered classes giving twenty-five verdict rows, because class 16 splits into two limbs that take different verdicts (16a global HAT inhibition is EXCLUDED, 16b selective CBP/p300 inhibition is NOT TESTED). Detail and sources in Appendix B.

#	Mechanism class	Rationale source	Direction evidence	DepMap on named targets	Verdict
1	Class I HDAC inhibition	PMID 28985013	AGAINST on six primary sources	HDAC3 0.874 flagged; HDAC1, HDAC2, HDAC4 to HDAC11 unflagged	**EXCLUDED** (Section 8.2)
2	PP2A activation	PMID 17717597	AGAINST on direction and exposure, two chemically unrelated molecules	PPP2CA 0.826, PPP2R1A 0.779 flagged; all five B56 subunits unflagged	**EXCLUDED** (Section 8.1)
3	APC/C inhibition	PMID 19407811	TOWARD, not excluded on direction	every subunit flagged, most at the ceiling	**EXCLUDED** by our dependency rule, not by a direction test (Section 8.3)
4	mTOR inhibition	screen only	direction question asked and empty, 10 records	MTOR 0.986 flagged	**EXCLUDED** (dependency rule alone)
5	Aurora B inhibition	PMID 12719470	AGAINST on both limbs at once	AURKB 1.000, blood median -2.420	**EXCLUDED**

#	Mechanism class	Rationale source	Direction evidence	DepMap on named targets	Verdict
6	PLK4, KIF18A and PLK1 inhibition	aneuploidy tolerance	AGAINST, PMID 26685311	PLK1 0.999, PLK4 0.982, KIF18A 0.885	**EXCLUDED** on dependency and direction
7	TTK/MPS1 inhibition	checkpoint kinase	AGAINST, PMID 29378962, PMID 25063032	TTK 0.948 flagged	**EXCLUDED** on direction
8	Sirtuin inhibition	PMID 24825348, inverted	AGAINST, PMID 24825348 verbatim; second molecule PMID 17355872	no sirtuin isoform flagged	**EXCLUDED** on direction
9	Sirtuin activation, SIRT2 K668 route	PMID 24825348	CONFLICTED, PMID 25285631 against PMID 29067790	as above, none flagged	**OPEN**
10	Transcriptional upregulation of *BUB1B* via SIRT1	PMID 35639235	TOWARD, one closed-access source, no corroboration and no contradiction	SIRT1 0.004, not flagged	**OPEN**, and the single most consequential unverified claim in the screen
11	NAD+ precursor supply	PMID 24825348	TOWARD in vivo mouse; no human demonstration concentration	NAMPT 0.296 flagged; entry enzymes unflagged	**OPEN** (Section 7.4)
12	Proteasome inhibition	PMID 19407811, PMID 21441410	CONFLICTED, PMID 21441410 against PMID 16432160	PSMB1 0.997, PSMB2 1.000, PSMB5 0.947, all flagged	**OPEN**
13	HSP90 inhibition	none located	one record, combination only, phospho endpoint	HSP90AA1 0.052, HSP90AB1 0.124, neither flagged	**NOT TESTED**
14	Chaperone induction or co-induction (HSF1)	PMID 30496702	TOWARD for HSPH1; part of the universe inhibits HSF1	HSF1 0.168, HSPH1 0.002, HSPA4 0.010, HSPB1 0.000 unflagged; HSPA8 0.864, HSPA5 0.993 flagged	**OPEN** as the screen scored it. **Our own reading of the same source after the citation fact-check is stricter and carries this arm at NOT TESTED** (Section 7.3)
15	Chemical chaperones	PMID 10677536, a different protein	**none; zero records pair the class with BUBR1**	HDAC3 0.874 flagged, which is 4-PBA's other activity	**NOT TESTED** (Section 7.2)
16a	Global lysine acetyltransferase inhibition, PCAF included	PMID 19407811	AGAINST, verbatim; the protective mark is the one removed	EP300 0.284 flagged; CREBBP 0.076, KAT2B 0.000	**EXCLUDED** on direction
16b	Selective CBP/p300 inhibition, sparing PCAF	PMID 24825348	no source measures BUBR1 after CBP/p300 HAT inhibition; exposure 37 to 184-fold short	as above	**NOT TESTED**, with the exposure arithmetic against it
17	Microtubule-targeting agents	co-occurrence only	AGAINST by two failure modes, PMID 24670687 and PMID 32850387, PMID 32764382	TUBA1B 0.997, TUBB 0.964, TUBB4B 0.262, all flagged	**EXCLUDED** on direction, with a gap in our own rule disclosed
18	DNA-damaging and topoisomerase cytotoxics	co-occurrence only	AGAINST, class source PMID 16449973 plus per-drug chromosome records	TOP2A 0.997, TOP1 0.979, both flagged	**EXCLUDED** on direction

#	Mechanism class	Rationale source	Direction evidence	DepMap on named targets	Verdict
19	Premature-termination-codon rescue	PMID 16411201	none located, and no approved read-through medicine currently exists	UPF1 0.998, UPF2 0.997, SMG1 0.946 (reduced denominator, see Table 2b), all flagged	**NOT TESTED**, both limbs negative for different reasons
20	Transcriptional upregulation of *BUB1B*, CDK5RAP2 route	PMID 19282672, PMID 23242215	none for any approved molecule	FOXM1 0.206 unflagged	**NOT TESTED**
21	mRNA or gene delivery of *BUB1B*	PMID 23242215	TOWARD at the genetic level	no drug target	**OPEN**, research stage, outside the approved-medicine scope
22	Senolytic clearance of senescent cells	PMID 22048312, PMID 16476774	addresses a different endpoint; one component excluded on direction, PMID 19395653	BCL2L1 0.846 flagged; BCL2 0.028, BCL2L2 0.019	**OPEN** for a downstream endpoint, not for raising BUBR1
23	Autophagy modulation	PMID 37094517	induction AGAINST, inhibition TOWARD, one source each way	ATG5 0.001, ATG7 0.001, ULK1 0.000, none flagged	**OPEN** on the inhibition limb only
24	Reduction of oxidative stress (antioxidants)	PMID 39845481	one segregation endpoint moved the right way; BUBR1 never measured after the agent	no named drug target	**NOT TESTED**

Totals over the 25 rows above: 11 EXCLUDED, 6 NOT TESTED, 8 OPEN.

The counting rule. A class counts as OPEN when a published rationale exists and no source excludes it. That is a test on the evidence, not on the endpoint, so a class stays OPEN even when the question it answers is not the one this section asks. Two of the eight open rows carry exactly that scope qualifier: class 21 is outside the approved-medicine scope of this screen, and class 22 addresses a downstream endpoint rather than BUBR1 itself. Both remain OPEN, because neither is excluded by any source and the qualifier concerns which question they answer rather than the strength of the evidence against them. A reader interested only in the classes that bear directly on raising BUBR1 or restoring the checkpoint should therefore read **six** of the eight open rows, with classes 21 and 22 set to one side on scope. **No row is counted as excluded on scope grounds anywhere in this section.**

An open verdict is not an endorsement: it means no source closes the class, which is a statement about the record rather than about biology. The classes are not ranked against each other and no molecule is ordered against another.

15.4 What the screen changed in our own hypotheses. Four things, folded into Sections 7 and 8.

- **Against us.** Class 15 removed the published basis for the chemical-chaperone arm we were carrying: a fixed search returned zero records, so that arm is MECHANISM NOT TESTED (Section 7.2).
- **Also against us, after a later citation check.** Class 14 appeared to supply the heat-shock arm with a BubR1-specific rationale (PMID 30496702), and the previous revision moved that arm to OPEN on it. Read in full, the paper's proposed mechanism runs through Cdc20 and it reports BubR1's association with Cdc20 as unaffected, so we carry that arm at MECHANISM NOT TESTED while the screen's class row stays OPEN. The divergence is flagged in both places rather than resolved in our favour (Section 7.3).
- **A second, independent route into our own NAD+ class.** Class 10 is a SIRT1 transcriptional route, and it inherits none of the doubts attached to the SIRT2 route: not the K668 against K250 dispute, not

the single-laboratory dependence of the K668 result, and not our own DepMap null for SIRT2 against BUB1B. It is also the single most consequential unverified claim in the screen, and the reasons are set out in Appendix B class 10. It does not change any verdict in Section 7, and it is the cheapest open question in the document to settle.
- **A disclosed asymmetry in our own exclusions.** Class 3 against class 12, in Section 8.3.

15.5 Three patterns in this literature, stated once rather than per class.

- **The tool-compound pattern, which appears four times.** The only experiment linking a mechanism to BubR1 is repeatedly done with a laboratory compound rather than a medicine: class 1 with trichostatin A, class 12 with MG-132, class 19 with no agent at all, and a JAK axis reported with a pan-JAK tool compound (PMID 20703093). This cuts both ways and must be applied symmetrically: tool-compound evidence was good enough to exclude class 1 at class level, so it is good enough to keep class 12 open at class level, and it is never good enough for a verdict on a specific medicine.
- **Almost nothing in this universe has a source about the drug itself.** Across every slice evaluated, only a handful of molecules have a primary source measuring that molecule on BUBR1 or the checkpoint: arsenic trioxide (PMID 19397590), bleomycin (PMID 18723675), two vinca alkaloids, curcumin through its microtubule action, and low-dose doxorubicin. Everything else is a database annotation, a tool compound, or a sensitivity paper.
- **Expected names do not come with evidence.** Metformin is the most-expected repurposing name in this universe and has no published BUBR1-raising rationale at all. The only primary records touching BUBR1 point away from the endpoint and do so far above clinical exposure: PMID 35568294 reports that "Metformin deregulated the expression of BUBR1 and MAD2, two core genes of spindle assembly checkpoint (SAC)" and "significantly increased genotoxicity in non-cancer cells", at 1 to 16 mM. Because that is above ordinary therapeutic plasma concentrations the frozen rule does not score it AGAINST at clinical exposure; it is recorded as unresolved with the AGAINST arm carried as a risk, not as a verdict either way.

15.6 What this section does not establish. It does not establish that any open class works. The counter-evidence behind most exclusions comes from cancer cell lines and mouse genetics at concentrations chosen to block a target completely, none of it in *BUB1B*-mutant or patient-derived cells, so it establishes the direction of a class and not a quantitative prediction. Every DepMap number is a loss-of-function proxy in cancer lines, never normal marrow, and for several classes the knockout direction is opposite to a drug's direction, so a flag is a measurement and not a verdict. Four limits of the screen are carried into the open and are repeated in Section 13: one slice of about 45 proteostasis and chaperone molecules was never evaluated drug by drug, so classes 13, 14 and 15 carry a coverage gap as well as an evidence gap, and two of those three classes are our own carried arms; the frozen direction rule has no limb for an agent that engages the checkpoint and still causes missegregation; the frozen exposure rule gives no tie-break between two verified demonstration concentrations, which left class 16b on the boundary between two outcomes; and the full text of the single source behind class 10 could not be obtained by any route available to this work. Nothing here has been tested in cells from a person with MVA, and none of it is a treatment recommendation.

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16. Scope of claims

This report makes eight claims, each backed by a result above.

1. The missense allele sits on the destabilising side of a calibrated abundance cut, with weak evidence that does not establish the abundance premise (Section 3).
2. The PP2A-activation mechanism is excluded on direction and exposure, using sources this report itself cites (Section 8.1).

3. The class I HDAC mechanism is excluded on direction at the class level, and no approved member enters (Section 8.2). The corollary is that one open hypothesis carries that activity as a liability.
4. For five of seven mechanism arms the exposure-versus-engagement arithmetic cannot be done at all, so the surviving arms are not ordered by engagement evidence (Section 7.1).
5. Criterion C4 does not measure the quantity it reasons about, so neither its exclusion nor its passes carry the weight previously placed on them (Section 9.3).
6. SIRT2 loss does not track BUB1B loss in cancer lines, at a threshold where the test had power 1.000 against the effect it pre-specified; the co-dependency screen's negative is a property of its rule (Section 9.4).
7. The NMN and NR human literature contains no *BUB1B*, SIRT2 or MVA population (Table 5).
8. Of twenty-five mechanism-class verdict rows, 11 are excluded by published primary sources or by a frozen test of ours, 6 have never been tested against BUBR1 at all, and 8 remain open, six of those bearing directly on raising BUBR1 or restoring the checkpoint; the literature pairing *BUB1B* with drugs is mostly about the checkpoint predicting cytotoxic sensitivity rather than about raising BUBR1; and a large share of the database-derived rows rest on target claims no primary source supports (Section 15).

It does not claim that any approved medicine treats or slows MVA. It does not rank any molecule above any other. Every mechanism here is a hypothesis for laboratory testing, which is what Track 2 asks for.

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17. Methods description (organisers' template)

The Track 2 submit form accepts exactly one file, so the organisers' methods answers cannot be uploaded separately and are reproduced here. Questions are quoted from `methods_description_form.xlsx`, sheet "Track 2 methods", rows 7 to 17 (sha256 `61aab080a286`, read from the sheet XML). Answers are deliberately short and cross-reference the body rather than repeating it.

Q1. Team name. DETA (detahub.com).

Q2. "Please describe your approach in detail (how you went from variant/mechanism to candidate medication(s))." We did not go from variant to medication. The evaluation chain is genotype-agnostic by design (Section 5): every drug and supplement is scored against the published mechanism, and verdicts are about mechanisms and evidence, never about a person. The chain has four stages. First, a structural gate: a pre-registered abundance-class prediction for the missense allele class, reported as weak evidence, which conditions every abundance-raising mechanism (Section 3). Second, mechanism-class enumeration: a pre-registered two-arm screen over a fixed 46-symbol gene set, one arm over drug-target databases and one over the literature, grouped into mechanism classes and scored EXCLUDED, NOT TESTED or OPEN against the published record (Section 15). Third, per-mechanism instruments applied symmetrically: on-target knockout dependency in DepMap 24Q4, a symmetric exposure and target-engagement test, and a mechanism-direction criterion that was itself validity-tested and found non-discriminating (Section 9, with the validity tests in Sections 9.3 and 9.4). Fourth, falsifiers: each surviving mechanism carries the experiment that would decide it (Section 10). Nine pre-registered tests are logged with hashes in Section 12. No ordinal ranking of molecules is produced, by decision (Section 1).

Q3. (required) "If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description." Generative AI was used. All entries are user-confirmed and not independently verified; the same disclosure appears in the front matter.

- **Claude, via Claude Code** (Anthropic, Claude Max consumer plan, training off, "Help improve Claude" off, no ratings or feedback given). Used for analysis code, running local analyses, and drafting. These sessions saw child-derived data, on local machines only. Model ids measured from the session transcripts: `claude-haiku-5-5`, `claude-opus-5-5`, `claude-sonnet-5-5`, `claude-opus-5`.
- **ChatGPT with Codex, "Sol"** (OpenAI, ChatGPT consumer app, training off, no ratings given). Used for

review and research. Model ids measured from the logs: `gpt-6-luna` in the box session of 2026-10-10 and `gpt-6.1-sol` in the two judge runs of 2026-10-18.
- **Gemini 3.1 Pro Deep Research** (Google, business account, data use off, no ratings given). Used for literature research.
- **Perplexity** (Perplexity AI), **Grok** (xAI) and **DeepSeek V4** (DeepSeek). Used for general background research and outside critique; no project data was shared with them; plan and data-handling settings were not recorded, and we do not invent them. No content was flagged for safety review for any tool. Every computed number in this report comes from a script whose inputs and decision rules were hashed before it ran, with the exceptions disclosed in Section 12.1.

Q4. "Was candidate identification automated (e.g., via a drug-target database query or LLM-assisted search), or primarily manual literature review and expert curation?" Both, in that order, and the boundary is stated. Enumeration was automated and pre-registered: DGIdb GraphQL over a fixed gene set keeping approved drugs only, ChEMBL by UniProt accession, four verbatim PubMed queries with names extracted against the Drugs@FDA bulk lexicon, plus scripted openFDA, ClinicalTrials.gov and DepMap queries. Adjudication was manual: every verdict rests on a source read by a person or agent and quoted verbatim, with abstract-level and full-text-level verification marked separately throughout. Language models assisted searching and drafting; no verdict rests on model output.

Q5. "Please describe any manual literature review or expert curation in detail." Sources were read and quoted verbatim, with the verification level marked at every use ("verified in full text", "verified at abstract", "verified at record level", or "unverified" with the reason). Two independent citation audits were run, the second re-deriving from primary sources before seeing the first; one cited source was found not to support the sentence it was attached to and that claim was withdrawn (Section 12). One result, the structural gate, was additionally subjected to an adversarial audit that reproduced it through an independent code path and then weakened the claim built on it (Section 3). Where a decisive full text was unreachable we say so and name the routes that failed rather than citing a review that quotes it. Where an earlier draft mischaracterised a source against our own interest, the correction is recorded (Section 4.5).

Q6. "Please state whether you relied solely on publicly available drug/target databases and literature (preferred), or whether proprietary data sources were also used." Solely public sources. No proprietary data source was used.

Q7. "Please describe the public data sources used in detail (e.g., DrugBank, ChEMBL, ClinicalTrials.gov, OMIM, PubMed literature search)." DepMap 24Q4 (CRISPRGeneEffect.csv, Model.csv); DGIdb; ChEMBL; PubMed and PubMed Central via E-utilities, plus Europe PMC; openFDA (drug label and Drugs@FDA endpoints) and DailyMed; the EMA's public medicine pages; ClinicalTrials.gov API v2; Human Protein Atlas; PubChem; UniProt; InterPro; AlphaFold DB; MaveDB. Exact queries, filters and dates are recorded with each result in Sections 9, 15 and Appendix B. Only gene, protein, drug and mechanism terms, PMIDs, DOIs and public accessions were sent to any external service.

Q8. "Please describe any proprietary data sources used in detail." None.

Q9. "How did you characterize the variant's mechanism (e.g., loss-of-function vs. gain-of-function, pathway disrupted, downstream biological consequence)?" By genotype class only: one truncating and one missense *BUB1B* allele, with the trans configuration **inferred, not measured**, because no sequencing template spans the gap between the two sites (Section 5). The pathway is the spindle assembly checkpoint and kinetochore-microtubule attachment, where BUBR1 abundance is dose-dependent and premature chromatid separation appears before aneuploidy (Section 4.2). The truncating allele is treated as a predicted nonsense-mediated-decay null. For the missense allele the question asked was narrower than loss of function: is it **abundance-limited or function-defective?** A pre-registered structural test places it on the destabilising side of a calibrated cut, with weak evidence that does not establish the premise, and that question is the report's gate rather than a conclusion (Section 3). The consequence is stated as a falsifier: if the allele is function-defective rather than abundance-limited, raising protein level could be

useless or harmful.

Q10. "Please provide an estimate of time or effort spent on this analysis (if possible)." Partially available, and we state what is missing rather than estimating it. A compute-cost audit covering the whole project measured 6.14 hours of serial wall clock and 1.74 GPU-hours on one RTX 3090 for the pipeline that produced the submitted results, and 25.1 hours serial with 4.10 GPU-hours including cross-checks and exploratory runs, at roughly 2.3 kWh. No rented or cloud compute was used. **The Track 2 share of that total was not separately measured**, and Track 2's work is overwhelmingly literature and database retrieval rather than compute, so the figures above are dominated by Track 1. Agent session time is logged in the register but was not apportioned per track. This is recorded as an open item rather than filled with a guess.

Q11. "Provide a method abstract (up to 500 words). Please include strengths and limitations of your approach as well as methodological detail." Use the **Abstract.** paragraph in this report's front matter, which is 445 words by whitespace count and therefore inside the template's 500-word cap for that spreadsheet field. Its methodological detail is the pre-registration ledger of twenty-one entries and their instruments; its stated strengths are pre-registration with hashes fixed before each run, with the exceptions disclosed in Section 12.1, symmetric application of every criterion, adversarial verification of our own strongest result, and negative results reported in full. Its stated limitations, set out in Section 13, are that all mechanism evidence is cell-based, mouse-based or predicted with none in cells from a person with MVA; that five of seven exposure arms have no establishable engagement concentration; that one of our own criteria is non-discriminating by its own validity test; that cancer risk, the central safety axis for this phenotype, has not been assessed for any mechanism; and that three limits of our own instruments are disclosed, namely a coverage gap in the molecule screen, a direction rule with no limb for an agent that engages the checkpoint and still causes missegregation, and an exposure rule with no tie-break between two verified demonstration concentrations.

Acknowledgements

The challenge rules require the following acknowledgement in any publication or public communication arising from participation, and it is reproduced verbatim:

> "This work was made possible through the Hackathon, organized by Sage Bionetworks in partnership with the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science), with prize sponsorship from AWS and Anthropic. We are deeply grateful to the child and their family who generously contributed their data and their story to advance research into this rare disease. We acknowledge their trust in making this Hackathon possible."

Dataset citation. The rules require the dataset to be cited using the reference provided on the challenge's Synapse page at the time of publication. No such reference was retrievable from the material available to us, so the dataset is cited here by its repository identifier: SageBio/mva-hackathon-2026-data at commit `59e322d`, accessed 2026-10-17. If the Synapse reference is published, it supersedes this line. No data from that dataset appears in this report, which works from genotype class only (Section 5).

Appendix A. Audit and correction history

Kept for auditability and moved out of the body deliberately. The hypothesis, not the correction record, is the deliverable.

A.1 The ranking rule and its withdrawal. Rule text, fixed before any candidate was ranked and never changed: candidates ordered by grade, ties resolved by fewer broadly-essential targets, and a supplement

or OTC drug sits above a prescription drug only on a strictly higher grade. An earlier draft produced an order the rule does not generate, and also asserted a clause ("no supplement is above an OTC drug") that the rule text does not contain; that application statement is withdrawn. The order was corrected under `APPLICATION-FIX-T2-01` with the rule text and every grade unchanged. Three further problems were then identified and are the reason the ordinal output is now withdrawn entirely: every prescription arm is G4, so the grade clause did no work and the whole order was set by a dependency flag count that this report's own safety section disowns as a proxy; the rule bars a supplement from sitting *above* a prescription drug but does not bar a tie, so the old rank 5 was reachable only through "G3-contested", which is not one of the four defined grades; and the tie-break counted flags with no denominator, so having fewer measurable targets improved an arm's position. The rule remains in the ledger as a record. It no longer orders anything.

A.2 Corrections carried from the previous revision. The withdrawn OTC clause; regulatory status no longer cited to a PMID; niacin's entry route corrected to NAPRT; PMID 24825348 identified as NMN-only, with its in vivo and MEF arms separated; NAMPT reported as flagged rather than as an unqualified "essential in blood lines"; the invented "fraction above 0.5" stop rule withdrawn; "targets that cannot be shown safe in cancer lines are not carried forward" corrected to what the flag actually supports; the blood-lineage claim matched to what was measured, with the two unmeasured genes named; ABL1 and SRC relabelled as comparators; paediatric labels reported per molecule; experiment IDs defined and traceable; three unresolved citation placeholders resolved to verified PMIDs; the trial count given its status filter; the read-through exclusion restated as a reasoned exclusion rather than a test; "biallelic loss" corrected to truncating plus missense throughout; K668 and K250 stated as opposite-sign assignments at different residues.

A.3 Review record. The previous revision was read by an adversarial simulated panel of seven reviews (four seats, three of them run twice as blind replicates, plus a second-provider reviewer at maximum effort). Each seat that was replicated reproduced its own assessment closely, so what the panel returned is a property of the report rather than of the draw. The panel's unanimous findings were that the ranked list was an artifact of a disowned proxy, that no exposure or engagement arithmetic existed, that criterion C4 had been applied asymmetrically, that the gate question undercut every ranked arm with nothing downstream changing, and that the epistemic apparatus was the strongest part of the document and should not be trimmed. The tests in Sections 3, 7.1, 8.2, 9.3 and 9.4 were pre-registered in response and run; two of them removed material the panel had suggested adding, and three of them weakened our own prior results. Five of seven reviews independently caught the omitted anemia observation now in Section 11.2, which is why that item is reported with its source sentence verbatim.

A.4 What changed in this revision (v4, 2026-10-20).

Edited in place. Nothing was deleted from the evidence files or the register. This record sits in the appendix, and not in the body, because three reviewers of the previous revision independently warned that the correction narrative had become more memorable than the hypothesis.

Framing. The ordinal list of drugs is withdrawn (user decision, 2026-10-20). Every entry is now a mechanism hypothesis with the test that kept it and the experiment that would decide it. Two of the new tests independently support the withdrawal: the engagement test finds no basis to order the survivors, and the grade that produced the old rank 5 is a contested label rather than one of the four defined grades.

New evidence incorporated. Five pre-registered tests with results: the structural gate and its adversarial verification (Section 3); the symmetric exposure and engagement test, which excludes the PP2A mechanism and reports two negative results (Section 7.1); the class I HDAC mechanism-class screen, zero entrants (Section 8.2); the C4 validity test, non-discriminating (Section 9.3); and the DepMap power test, which keeps one null and relabels the other (Section 9.4).

New sections. Clinical phenotype and what an intervention could touch (Section 2), including locus heterogeneity and *TRIP13* labelled as an MVA gene. The gate, moved to the front (Section 3). Real label

warnings, the anemia observation from our own cited source, and cancer risk as a required test (Section 11). A plain-language section for families (Section 14). The Non-choices mechanism-class screen (Section 15), with its per-class detail in Appendix B and its new dependency measurements in Table 2b.

Two of our own arms moved by the class screen, both downwards. Sodium phenylbutyrate's chemical-chaperone arm is downgraded to MECHANISM NOT TESTED, because a fixed search pairing BUBR1 with chemical chaperones, phenylbutyrate, TUDCA, ambroxol and trehalose returned zero records; with its HDAC arm excluded on direction and reached at clinical concentration, neither of its arms now has a source connecting it to BUBR1 in the direction we need, and we say so rather than keep it on a rationale the search could not find. Arimoclomol's arm stays at MECHANISM NOT TESTED: the class screen scored its class OPEN on PMID 30496702, and a later citation check found that the paper's proposed mechanism runs through Cdc20 and that it reports BubR1's association with Cdc20 as unaffected, so the upgrade was reversed rather than kept (Section 7.3). The deciding experiment is restated as whether a co-inducer raises HSPH1 and BUBR1 in unstressed cells. The APC/C against proteasome asymmetry in our own exclusions is disclosed in the body (Section 8.3), not only in an appendix.

Second refresh of the class screen (same day). The screen was rerun and extended from 21 classes to twenty-four numbered classes giving twenty-five verdict rows, because class 16 splits into two limbs that take different verdicts. Three settlements were folded in. Curcumin closes as MECHANISM NOT TESTED on three independent grounds, including the trap that the one paper measuring BubR1 after curcumin is a microtubule experiment in which BubR1 rises **because the drug broke the spindle**, which would have been wrong to cite as support. Vitamin C closes as MECHANISM NOT TESTED because BUBR1 was never measured after the agent and N-acetylcysteine reproduced the effect at the same concentration, so the active principle reads as antioxidant capacity rather than ascorbate. Both are reported in full because they are sold in every shop. A SIRT1 transcriptional route is separated out as its own class and flagged as the single most consequential unverified claim in the screen, with its full text unreachable and transcript against protein not established. Three limits of our own instruments are now disclosed in Section 13: a coverage gap in the molecule universe that falls on two of our own arms, a direction rule with no limb for an agent that engages the checkpoint and still causes missegregation, and an exposure rule with no tie-break between two verified demonstration concentrations. A counting discrepancy in the screen was found and settled: two prose sentences had said twelve excluded, six not tested and seven open over twenty-four, while the table labelled eleven, six and eight over twenty-five rows. The table was correct, the two sentences were corrected, and **no verdict and no number changed**. In particular, senolytic clearance stays OPEN: it has a published rationale, no source excludes it, and a deciding experiment is stated, so relabelling it would have been a verdict change dressed as a labelling fix. The screen now states its counting rule directly under the table, and Section 15.3 mirrors it rather than paraphrasing it.

Corrections. PMID 25672491 is a 19-author paper reporting independent pharmacological support for the SIRT2 axis, not a same-laboratory echo; the single-laboratory claim is withdrawn and the shared-author fact stated accurately (Section 4.5). The PP2A-B56 claim is verified and its "unverified" tag retired (Section 4.6). PMID 18932004's PCS finding is reported, which raises the effective target above 50% residual (Section 4.2). BUB1B's own dependency and variance are reported (Table 2). The flag count now carries a denominator (Table 3). The 1.25-fold go/no-go threshold is withdrawn as too small to mean anything on this report's own anchor (Section 10). The falsifier set is rebuilt so that arms separate rather than collapse, including a dominant-negative falsifier that did not exist. The 0.135 and 0.358 uM figures are identified as nicotinamide, not NMN or NR (Section 12.1 item 10). The data-handling disclosure and the reproducibility statement now agree.

Appendix B. The mechanism classes, one paragraph each

Detail behind the summary table in Section 15.3. Twenty-four numbered classes giving twenty-five verdict rows, because class 16 splits into two limbs that take different verdicts. VERIFIED means retrieved from PubMed E-utilities, from a regulator's own page, or computed from the public DepMap 24Q4 files during

the screen; anything else is labelled inline. Classes treated at length in the body are summarised here only.

1. Class I HDAC inhibition. EXCLUDED. In full in Section 8.2. Rationale PMID 28985013. The verdict rests on three sources: trichostatin A abolishing BubR1 phosphorylation and kinetochore localisation after spindle poison (PMID 15753652), transcription-independent checkpoint override (PMID 17616695), and aneugenicity in human lymphoblasts by nondisjunction (PMID 16857700). Three further sources are cited narrowly, each with a stated limit, and the verdict does not depend on them (Section 8.2). The deciding point is that the process the lead describes is the normal termination of checkpoint signalling, so blocking it gives a mitotic block rather than a better checkpoint.

2. PP2A activation. EXCLUDED, and the verdict is now known not to be about one molecule. In full in Section 8.1. New from the class screen: a second, chemically unrelated molecule fell to the same ruling. Maduramicin's universe claim was that it targets PPP2CA, and its primary source resolves the sign of that interaction as activation (PMID 29241734, title verbatim: "Maduramicin-activated protein phosphatase 2A results in extracellular signal-regulated kinase 1/2 inhibition, leading to cytotoxicity in myocardial H9c2 cells"; the paper carries an erratum, PMID 30213496, which we have not read); it has no authorised human drug product and is a veterinary anticoccidial. Two molecules with nothing in common but the mechanism is better evidence for a class verdict than one. Also new: the five PP2A B56 regulatory subunits are all unflagged (PPP2R5A 0.000 to PPP2R5C 0.036) while the catalytic and scaffold subunits are flagged, so the dependency signal sits on the core holoenzyme rather than on the subunits that do the BUBR1-directed work.

3. APC/C inhibition. EXCLUDED by our frozen dependency rule, and this is the one exclusion whose direction evidence points the other way. PMID 19407811 (Choi 2009, *EMBO J*) establishes a good rationale: "APC/C-Cdc20 was responsible for BubR1 degradation during mitosis", so inhibiting APC/C should raise BubR1. It was removed by `T2-P4` on dependency, and every subunit measured is flagged, most at or near the ceiling (CDC20 1.000, ANAPC1 1.000, ANAPC5 1.000, CDC16 1.000, CDC23 1.000, CDC27 1.000, ANAPC11 0.991, ANAPC2 0.977, UBE2C 0.767, UBE2S 0.292). The asymmetry against class 12 is disclosed in Section 8.3.

4. mTOR inhibition. EXCLUDED by `T2-P4`, MTOR 0.986, blood median -1.130. The direction question was asked and came back empty: a query pairing BubR1 or *BUB1B* with mTOR, rapamycin, everolimus or sirolimus returns 10 records, all of them BUBR1-drives-mTOR-signalling or biomarker papers, none showing a rapalog raising BUBR1. The exclusion rests on the dependency rule alone and not on a direction finding, and is recorded that way rather than dressed up. Two further molecules took the class disposal on their own pharmacological class and then turned out to have no authorised drug product in any register queried, which is the admission-ticket failure the pre-registration said the approval flags could produce.

5. Aurora B inhibition. EXCLUDED, cleanly, on both limbs. Class source PMID 12719470 (Ditchfield 2003, *J Cell Biol*), verbatim on the Aurora kinase inhibitor ZM447439: "ZM447439-treated cells exit mitosis with normal kinetics, indicating that the spindle checkpoint is compromised", "ZM447439 prevents mitotic arrest after exposure to paclitaxel", and "In the absence of Aurora B function, kinetochore localization of the spindle checkpoint components BubR1, Mad2, and Cenp-E is diminished". Both halves of the frozen AGAINST definition are met at once: BUBR1 at kinetochores is lowered and the checkpoint is compromised. Supported by PMID 37443114 on "abrogation of the spindle assembly checkpoint by AURKB or TTK inhibitors" as a class effect. Five approved kinase inhibitors entered the universe on an AURKB claim and fall to it. AURKB is 1.000 with a blood-lineage median of -2.420. It is not the single most extreme dependency reading in this project on either metric: seven other genes also reach a fraction of 1.000, and PLK1 (-2.772) and CDC27 (-2.632) have lower blood-lineage medians. This class is also why the retired PP2A falsifier in Section 10 was one-sided in the wrong direction.

6. PLK4, KIF18A and PLK1 inhibition. EXCLUDED on dependency (PLK4 0.982, KIF18A 0.885) and also on direction. PMID 26685311 verbatim: "PLK1 inhibition, together with partial inhibition of Aurora B, allows efficient SAC override to occur". PLK1 is 0.999 with a blood median of -2.772. A class whose

documented effect is checkpoint override points away from the endpoint.

7. TTK/MPS1 inhibition. EXCLUDED on direction. MPS1 is the apical checkpoint kinase, so inhibiting it is the most direct available way to weaken what we want strengthened. PMID 29378962 verbatim, on a clinical TTK inhibitor: it "induced apoptosis and potentiated aneuploidy in TNBC lines by accelerating progression through mitosis and inducing mitotic segregation errors". PMID 25063032 adds the mirror image, that MPS1 overexpression also "increases genome instability in tumor cells owing to a weakened spindle assembly checkpoint", so the relation is not monotonic; neither reading supports inhibition. TTK 0.948, blood median -0.841.

8. Sirtuin inhibition. EXCLUDED on direction. This is the class our own removed molecule belongs to. The anchor, verbatim (PMID 24825348): "the loss of BubR1 levels with age is due to a decline in NAD(+) and the ability of SIRT2 to maintain lysine-668 of BubR1 in a deacetylated state, which is counteracted by the acetyltransferase CBP"; and the inhibitor experiment directly: "Treatment with either of these inhibitors resulted in a reduction in BubR1 protein levels", nicotinamide at 20 mM for 16 hours in HeLa cells. Reported honestly: 20 mM in culture is above the 1.28 mM mean peak plasma level at oral doses up to 6 g (PMID 8532910, pooled across arms), so that experiment establishes the sign and not the reachability, and Section 9.3 separately shows the plasma-versus-IC50 instrument cannot discriminate. A second molecule falls to the same ruling: suramin, which "inhibits SIRT5 NAD(+)-dependent deacetylase activity with an IC(50) value of 22 microM" (PMID 17355872). **Marked UNVERIFIED:** that demonstration is on SIRT5 and the step to SIRT2 is an isoform extrapolation, supported by the shared NAD+ pocket but not itself measured. No sirtuin isoform is flagged (SIRT3 0.003, SIRT6 0.000, SIRT7 0.033).

9. Sirtuin activation, the SIRT2 K668 route. OPEN, and conflicted. Two primary sources disagree about the sign of the step, at two different lysines. PMID 25285631 states that "acetylation of BubR1 at lysine 250 (K250) during prometaphase inhibits its APC/C-dependent proteolysis" and that "SIRT2 deacetylates BubR1 K250", making SIRT2 activity a route to less BubR1; the same paper reports that SIRT2 knockdown raised K250 acetylation but "this increase is not substantial to elevate the levels of total BubR1". Against that, PMID 29067790 in mouse oocytes found the non-acetylatable mutant BubR1-K243R "markedly suppresses spindle/chromosome anomalies and thereupon lowers the incidence of aneuploid eggs" in aged oocytes, which makes deacetylation protective. No approved or supplement molecule has ever been tested on this: the corresponding search returns **zero records**. **Deciding experiment:** in a human cell model with reduced *BUB1B* expression, measure total BubR1 and BubR1-K250 acetylation by quantitative immunoblot at a fixed mitotic index under a selective SIRT2 activator against vehicle, with a sirtuin-inhibitor arm as the sign control. If SIRT2 gain lowers total BubR1, the class closes.

10. Transcriptional upregulation of *BUB1B* via SIRT1. OPEN, and the single most consequential unverified claim in this screen. PMID 35639235 (Zhao et al, *Mol Cell Biochem* 2022) reports, verbatim: "ChIP analysis revealed that SIRT1 could bind to the promoter of BubR1, a component of spindle assembly checkpoint (SAC), to upregulate its expression. Overexpression of BubR1 decreased MC whereas knockdown of BubR1 increased it." It matters because it is a second and independent route into the NAD+ class: every doubt raised about that class so far attaches to SIRT2 specifically, and a SIRT1 transcriptional route inherits none of them. What was verified, and what was not, stated precisely.

- **Group independence: VERIFIED.** The authors are at the College of Life Sciences, Wuhan University, read from the Europe PMC record. They are independent of the author line that produced PMID 24825348.
- **System: VERIFIED at abstract.** Cancer cell lines, mitotic catastrophe induced pharmacologically, and the SIRT1 manipulation is **genetic** (overexpression and knockdown), not pharmacological.
- **Transcript or protein: NOT ESTABLISHED.** The abstract says "upregulate its expression" and does not say which. This is the question the full text would answer and the one that decides whether the route is usable.
- **Full-text verification FAILED, with the routes named.** The paper is not in PMC, is not open access in Europe PMC, and OpenAlex reports it closed with no repository copy. Unpaywall was not queried

because it requires sending a contact email, which this work's rules forbid. Nothing is claimed from the full text.
- **Corroboration: none, and no contradiction either.** A SIRT1 with *BUB1B* search returns 6 records and only this one makes the promoter claim; the rest are the K668 paper, a review noting that BubR1, YAP and SIRT1 all decline with age, and three unrelated records. The paper has 8 citations.
- DepMap: SIRT1 0.004, blood median +0.089, not flagged.

So the honest state is one closed-access paper, one group, no corroboration, no contradiction, a genetic rather than pharmacological manipulation, and an endpoint word that could mean transcript or protein. **Deciding experiment, and it is cheap:** in a human cell line with reduced *BUB1B* expression, measure *BUB1B* transcript by qPCR and total BubR1 protein by immunoblot after a selective SIRT1 activator against vehicle, with SIRT1 knockdown as the sign control, and repeat the promoter ChIP. What settles it is whether promoter occupancy translates into protein. Note the internal tension this paper also creates: it reports autophagy as protective against mitotic catastrophe, where class 23's source reports autophagy degrading BubR1. Both are single sources and they do not agree.

11. NAD+ precursor supply. OPEN. In full in Section 7.4. **Deciding experiment as the screen states it:** establish the concentration at which NMN or NR raises BubR1 protein in a human cell line, then compare it with measured intracellular NAD+ and nicotinamide in that same compartment rather than with a plasma figure, which Section 9.3 shows cannot discriminate.

12. Proteasome inhibition. OPEN, with conflicting direction evidence and a maximal dependency signal. The rationale follows from class 3: BubR1 is an APC/C substrate, so blocking the proteasome should stabilise it. PMID 21441410 supports exactly that, verbatim: mitotic slippage "was blocked by the proteasome inhibitor MG-132 and involved proteasome-dependent degradation of cyclin B1 and the mitotic checkpoint proteins budding uninhibited by benzimidazole related 1 (BubR1) and cell division cycle 20 (Cdc20)". PMID 16432160 points the other way: a dominant-negative 19S ATPase subunit that lowered measurable proteasome activity produced "a significant decrease in expression of BubR1". The two are not equivalent interventions and the conflict is real. Two things about the standard of evidence must be said together, because they cut in opposite directions. The MG-132 demonstration uses a tool compound, not a medicine, which is the same situation that left 4-PBA's HDAC arm unresolved at drug level, and the three approved proteasome inhibitors in the universe are correspondingly NOT EVALUABLE on two named failed searches. But at the level of a mechanism class, tool-compound evidence is admissible, and it is precisely tool-compound evidence that carried the class-level exclusion of class 1. Applying the same standard here leaves this class open and conflicted rather than excluded, and we apply it symmetrically rather than selectively. The catalytic subunits are at the dependency ceiling (PSMB1 0.997 blood median -1.804, PSMB2 1.000 blood median -2.050, PSMB5 0.947 blood median -0.939), and per the frozen rule a flag alone removes nothing here. **Deciding experiment:** in non-transformed human cells, at the lowest concentration giving measurable 20S inhibition, measure total BubR1, BubR1 kinetochore occupancy and the missegregation rate in the same cells, which separates "more BubR1 protein" from "a working checkpoint".

13. HSP90 inhibition. NOT TESTED. Missing datum: any measurement of BUBR1 protein level under HSP90 inhibition alone. Failed search routes: a query pairing BubR1 or *BUB1B* with Hsp90, geldanamycin or 17-AAG returned one record, and a second naming the paralogues and four named inhibitors returned two, one of them unrelated. The single relevant record (PMID 19371599) is a combination experiment in which 17-DMAG plus arsenic trioxide "significantly increased ATO-induced mitotic arrest and ATO-induced BUBR1 phosphorylation" while also significantly increasing abnormal mitotic spindles; phosphorylation is not protein level and no single-agent arm exists. Neither paralogue is flagged (HSP90AA1 0.052, HSP90AB1 0.124), so the obstacle is evidence, not dependency. Thirty-three rows of the universe entered tagged to these two genes, the largest single mechanism tag in the screen, and they sit in the one slice that was never evaluated drug by drug, so this class carries a coverage gap as well as an evidence gap.

14. Chaperone induction or co-induction through HSF1. OPEN as the screen scored it; we carry

the arm at NOT TESTED. Section 7.3 gives the reasoning: PMID 30496702's proposed mechanism runs through Cdc20, it reports that HSPH1 knockdown "did not affect coprecipitation of Mad2 and BubR1 with Cdc20", and its only BubR1-specific observation is loss of kinetochore localisation under heat shock. Counterweights inside the class: the approved co-inducer has no demonstrated engagement step at any concentration and a label that disclaims its mechanism, and a class tag is not a direction, since quercetin enters the same universe tagged to HSF1 and is used in the literature as an HSF1 and HSP70 inhibitor (PMID 24296267, PMID 23219797). This class also sits in the unevaluated proteostasis slice.

15. Chemical chaperones. NOT TESTED, and the search result is unusually clean. In Section 7.2. Missing datum: whether any BUBR1 substitution is a chaperone client at all. The fixed query returned **zero records**, and the class rationale is transferred from alpha1-antitrypsin Z (PMID 10677536). Phenylbutyrate is the one molecule in the whole screen whose achieved concentration is in the right order of magnitude, which makes the untested question more worth answering, not less, and it also carries the class 1 liability. **Deciding experiment:** a thermal-shift or limited-proteolysis stability assay on recombinant BUBR1 kinase-domain variants with and without 4-PBA at the millimolar concentrations the plasma figure supports, followed by total BubR1 protein in cells. This class sits in the unevaluated proteostasis slice.

16. Lysine acetyltransferase inhibition. The class splits on selectivity, and scoring it as one block would have got one of its two limbs wrong. Two acetyltransferases act on BubR1 at two lysines in opposite directions. CBP acetylates K668 and drives BubR1 loss: SIRT2 maintains "lysine-668 of BubR1 in a deacetylated state, which is counteracted by the acetyltransferase CBP" (PMID 24825348, verbatim). PCAF acetylates K250 and protects BubR1: "BubR1 acetylation was required for checkpoint function, through the inhibition of ubiquitin-dependent BubR1 degradation", and "the acetylation-deficient mutant, BubR1-K250R, was unstable" (PMID 19407811, verbatim). So inhibiting CBP or p300 points toward the aim, inhibiting PCAF points against it, and the verdict depends entirely on which enzymes a molecule hits.

16a. Global HAT inhibition, PCAF included. EXCLUDED on direction. The protective mark is the one that gets removed. PMID 22340495 shows it by removing the recruiter: "In the absence of BRCA2, BubR1 acetylation is abolished, and the level of BubR1 decreases during mitosis", with the resulting mice showing "weakened SAC activity, and aneuploidy". PMID 33955658 and PMID 25559578 confirm at organism level that acetylation-deficient BubR1 mice develop chromosome instability and spontaneous tumours. The global HAT inhibitor in the universe is epigallocatechin gallate, identified as "a novel HATi with global specificity for the majority of HAT enzymes" at 100 uM (PMID 19147572, verbatim), which is both counter-directional and far above any human free concentration. EP300 is flagged at 0.284 (blood median -0.761); CREBBP 0.076 and KAT2B 0.000 are not.

16b. Selective CBP/p300 inhibition, sparing PCAF. NOT TESTED, and the exposure arithmetic is against it. The one molecule occupying this limb is curcumin, selective by source: "a specific inhibitor of the p300/CREB-binding protein (CBP) HAT activity but not of p300/CBP-associated factor" (PMID 15383533, verbatim). On paper that is the right selectivity. It fails three separate checks, and this one is set out at length because curcumin is sold in every shop and because citing the wrong paper here would have been an easy mistake to make.

- *Direction, the missing datum.* **No source measures BUBR1 after inhibiting CBP or p300 HAT activity.** The fixed search returns 2 records. One is a transcriptomics paper in which *BUB1B* is a hub gene. The other, PMID 20646066, does measure BubR1 after curcumin, and it measures it through a completely different mechanism: "Brief treatment with curcumin depolymerized mitotic microtubules, perturbed microtubule-kinetochore attachment and disturbed the mitotic spindle structure", "induced monopolar spindle formation", and then "curcumin increased the accumulation of Mad2 and BubR1 at the kinetochores, indicating that it activated the mitotic checkpoint". **BubR1 rises at kinetochores because the drug broke the spindle.** That is the checkpoint doing its job against a lesion the drug caused, and reading it as support for a stabilisation route would have been wrong. It also places curcumin in class 17.
- *Exposure, settled both ways rather than conveniently.* Free curcumin is an upper bound, not a measurement:

at 10 g and 12 g oral with a 50 ng/mL limit of detection, "only one subject had detectable free curcumin at any of the 14 time points assayed" (PMID 18559556), so for 11 of 12 subjects free curcumin is below **0.136 uM**. The demonstrated concentration had to come from a different source, because the selectivity paper's full text was unreachable on three routes; the replacement is full-text verified: the p300-HAT-dependent effect is demonstrated with "curcumin (5 or 10 uM)" over 48 hours in primary neonatal rat cardiomyocytes (PMID 18292809, read at PMC2248328). Taking the lowest demonstrated concentration, the choice most favourable to the molecule, the ratio is below 0.027; at 10 uM it is below 0.014; at the 25 uM figure in PMID 17927689 it is below 0.0054. Under the frozen rule, failure on exposure needs a ratio below 0.01 and the weak band is 0.01 to 0.1, so **the arm is E-WEAK at the most generous demonstration concentration and E-FAIL at the least generous, and the frozen rule provides no tie-break between two verified demonstrations.** Both readings are reported rather than one chosen. Either way the margin is 37 to 184-fold short, against a free concentration that is itself only an upper bound.

- *Two counter-directional arms at cellular concentrations.* In Raji cells "significant decreases in the amounts of p300, HDAC1 and HDAC3 were detected after treatment with curcumin", with a 24-hour IC50 of 25 umol/L (PMID 17927689). That is the class I HDAC route, excluded on direction on six primary sources including aneugenicity in human cells. And PMID 20646066 above is the microtubule route, class 17.

So the molecule occupying the one favourable limb of this class carries, at comparable concentrations, one arm into an excluded class and a second arm into another excluded class, while the favourable arm itself has never been tested on BUBR1 and is two orders of magnitude out of reach. **Stated plainly because it is sold everywhere as a supplement: this does not read as open.**

17. Microtubule-targeting agents. EXCLUDED on direction, by two separate failure modes, and with a gap in our own rule disclosed. The largest co-occurrence block in the screen, present because the checkpoint predicts their cytotoxicity and not because anyone proposed them to raise BUBR1. The first failure mode is concentration-resolved: PMID 24670687 (Zasadil 2014, *Sci Transl Med*) measured the drug in human tumours and found "paclitaxel levels in primary breast tumors are well below those required to elicit sustained mitotic arrest. Instead, cells in these lower concentrations of drug proceed through mitosis without substantial delay and divide their chromosomes on multipolar spindles, resulting in chromosome missegregation"; PMID 37883329 generalises it across the class (that paper carries an erratum, PMID 39938018, which we have not read). The second is molecular and specific to the vinca alkaloids: vincristine-induced "phosphorylation and cleavage of BUBR1" (PMID 32850387, verbatim) and the same for vinorelbine (PMID 32764382), where cleavage is a net loss of intact BUBR1 and a different thing from checkpoint override; concentrations are UNVERIFIED in both abstracts. **The rule gap, reported as a limitation of our own instrument.** The frozen AGAINST definition is to lower BUBR1 protein, or to silence, weaken or override the checkpoint. Colchicine, and curcumin in PMID 20646066, do the opposite of the second limb: they **engage** the checkpoint and still cause missegregation, through a spindle geometry the checkpoint cannot sense. Those molecules therefore fall outside the frozen definition and stay unresolved individually, even though aneugenicity is the documented risk and aneuploidy is the disease endpoint. Scoring them AGAINST would mean reading the definition more loosely after seeing the drug, which is what pre-registration exists to prevent. The honest statement is that our direction rule has no limb for an agent that activates the checkpoint while still causing missegregation, and any fix must be a recorded amendment and not a change made inside an evaluation. TUBA1B 0.997, TUBB 0.964, TUBB4B 0.262.

18. DNA-damaging and topoisomerase cytotoxics. EXCLUDED on direction, and the way the verdict was reached is worth stating because it shows the rule working. One evaluation of this class found only a single bleomycin paper and correctly **refused** to stretch it across nine other cytotoxics, leaving them not evaluable. A second evaluation then found the class source the first could not. PMID 16449973 is BUBR1-specific and in the right setting: "Following treatment with doxorubicin, BubR1(+/-) murine fibroblast cells (MEF) were defective in undergoing G(2)/M arrest", and "whereas in the presence of DNA damage BubR1(+/+) MEF cells remained arrested in mitosis, BubR1(+/-) MEFs rapidly exited from mitosis

and divided". In BubR1 insufficiency, an agent whose approved action is to damage DNA is met by a cell that divides instead of arresting. On that class-level ground ten approved cytotoxics fall, supported by per-drug chromosome-level records, of which five carry the class limb verified and the per-drug limb UNVERIFIED as marked in the underlying evaluation. Protein-level confirmation comes from PMID 15870702, where chronic low-dose doxorubicin in human hepatoma lines produced a senescence-like phenotype "preceded by multinucleation and downregulation of multiple proteins with mitotic checkpoint function, including CENP-A, Mad2, BubR1, and Chk1", and PMID 19282672 gives a transcriptional route, since CDK5RAP2 "resides on the BUBR1 and MAD2 promoters and regulates their transcription" and "cancer cells cultured in the presence of paclitaxel or doxorubicin exhibit dramatically decreased CDK5RAP2 levels". Both halves belong in the record: the refusal to over-extend one paper, and the later finding of a proper class source. TOP2A 0.997 and TOP1 0.979 are both flagged.

19. Premature-termination-codon rescue: translational read-through and NMD inhibition. NOT TESTED, and both limbs are negative for different reasons. The rationale is stated in the primary MVA literature: "Transcripts derived from the patients with the 1833delT mutation and the splice site mutation were significantly reduced, probably due to nonsense-mediated mRNA decay" (PMID 16411201, verbatim), in a study that also shows restoring *BUB1B* by chromosome transfer "restored normal BubR1 levels, kinetochore localization of p55cdc, and the normal responses to colcemid treatment". A transcript-level loss route is documented for nonsense and splice alleles as a class, and a protein-level rescue is documented. The two drug limbs fail separately. **Designed read-through agents: there is currently no approved one.** The EMA medicine page for the one such product states verbatim that "On 28 March 2025, the European Commission issued a decision to not renew the conditional marketing authorisation for Translarna (ataluren) in the European Union (EU)", with "This medicine's authorisation has expired" and the committee ground that "the effectiveness of the medicine has not been confirmed"; openFDA returned zero results on both the label and the approvals endpoints, by generic and by substance name. That is a regulatory-status finding and not a claim about the molecule's biology, and it is a cleaner reason than any inference from mechanism. Separately, the discovery assay for that molecule carries a published contradiction (PMID 20194791: it "is an unusually potent FLuc-inhibitor"). **Aminoglycosides are approved**, and gentamicin is the only molecule in the entire universe whose mechanism matches the truncating-allele class, but the literature has nothing: the corresponding search returns **count 0**. No source establishes the sign of the effect on BUBR1 at any exposure and no demonstration concentration exists to compare against the label figures. The arm is unresolved, it is not removed, and absence of data is not read as safety. The NMD machinery is at the dependency ceiling (UPF1 0.998, UPF2 0.997, SMG1 0.946, the last on a reduced denominator of 349 lines rather than 1,178 and 22 blood lines rather than 135, as disclosed in Table 2b), which is a reason to ask the question carefully rather than a reason not to ask it. **Deciding experiment:** in a cell model carrying a *BUB1B* nonsense allele, measure full-length BUBR1 by immunoblot and *BUB1B* transcript by qPCR under a read-through agent and under NMD knockdown, against a matched model with no premature stop.

20. Transcriptional upregulation of *BUB1B*, the CDK5RAP2 route. NOT TESTED. The aim is the right one. PMID 23242215 (Baker 2013, *Nat Cell Biol*) showed that "sustained high-level expression of BubR1 preserves genomic integrity and reduces tumorigenesis", acting "by correcting mitotic checkpoint impairment and microtubule-kinetochore attachment defects", and that it "extends lifespan and delays age-related deterioration and aneuploidy in several tissues". A named transcriptional regulator exists (CDK5RAP2, PMID 19282672). Missing datum: any approved molecule shown to raise *BUB1B* transcription through it. The failed search returned 23 records, all descriptive expression or tumour-marker studies and none a drug intervention. FOXM1 is unflagged at 0.206. Class 10 is the one transcriptional route in this screen with a named upstream regulator a drug could plausibly engage.

21. mRNA or gene delivery of *BUB1B*. OPEN, and explicitly outside the approved-medicine scope of the screen. PMID 23242215 is the strongest single piece of evidence anywhere in this report that more BUBR1 is the right target, because it is the only source that manipulates the quantity directly and measures aneuploidy and lifespan. It is a transgenic mouse, not a medicine. The screen's inclusion rule admits only

approved drugs and supplements, so nothing in this class could have entered the universe; that is a property of the procedure, not a judgement about the approach, and it is recorded so the absence is not mistaken for a negative result. **Deciding experiment:** dose-controlled *BUB1B* expression in a human cell model with reduced BUBR1, measuring the missegregation rate against expression level, which also establishes how much of a rise is enough. This is the class our own research-stage mRNA idea (Section 7.5) belongs to.

22. Senolytic clearance of senescent cells. OPEN for a different endpoint. The rationale uses the BubR1 model directly. PMID 22048312 (Baker 2011, *Nature*) verbatim: "in the BubR1 progeroid mouse background, INK-ATTAC removes p16(Ink4a)-positive senescent cells upon drug treatment", and "life-long removal of p16(Ink4a)-expressing cells delayed onset of these phenotypes", with late-life clearance attenuating already-established disorders. PMID 16476774 supports the premise that the phenotypes track senescence rather than aneuploidy itself. The honest statement is that this class does not raise BUBR1 and does not restore the checkpoint: it addresses a downstream consequence, so it is outside the question this report asks and should not be scored against the classes that answer it. One senolytic component is separately excluded on direction: "in unperturbed human cancer cells fisetin caused premature initiation of chromosome segregation and exit from mitosis without normal cytokinesis … Aurora B, Bub1, BubR1 and Cenp-F rapidly lost their kinetochore/centromere localization" (PMID 19395653, verbatim); because that is reported in unperturbed cells, the stress-dependence clause does not rescue it. BCL2L1 is flagged at 0.846; BCL2 0.028 and BCL2L2 0.019 are not. **Deciding experiment:** measure senescent burden, not BUBR1, as the endpoint in a *BUB1B*-low human cell model.

23. Autophagy modulation. OPEN on the inhibition limb only, and the two limbs point opposite ways. One source addresses it directly. PMID 37094517 verbatim: "there is an initial increase in Bub1b/BubR1 levels, feasibly as part of the cells' response against OS-driven genomic instability, that is followed by its autophagy dependent degradation", with the paper also noting that "the proteasome function decays as cells age". If autophagy degrades BubR1, autophagy inducers in the universe point against the mechanism and autophagy inhibitors point toward it. One source is not a class verdict, this one is in an oxidative-stress senescence model rather than a checkpoint-deficiency model, and class 10's source reports autophagy as protective against mitotic catastrophe, which does not agree with it. ATG5 0.001, ATG7 0.001, ULK1 0.000, none flagged. **Deciding experiment:** a cycloheximide-chase measurement of BubR1 half-life in a *BUB1B*-low human line with an autophagy inhibitor, a proteasome inhibitor and vehicle in the same experiment, which settles which degradation route dominates and therefore which limb is worth pursuing.

24. Reduction of oxidative stress. NOT TESTED, and it is the only class in which an agent moved a segregation endpoint in the right direction. PMID 39845481 (Nago et al, *Reprod Med Biol* 2025, full text read at PMC11751902) reports that in Sod1-deficient mouse oocytes matured in vitro at 21% oxygen, chromosome misalignment and segregation errors rose, and "These changes were associated with a decrease in the recruitment of BUBR1 to kinetochores at M I stage"; then "the addition of L-ascorbic acid (AsA) or N-acetyl-L-cysteine (NAC) during IVM reduced the frequency of chromosome segregation errors in Sod1 KO oocytes", at **500 uM** of either agent, with the rate of correctly-numbered oocytes raised and premature sister chromatid separation lowered. That is a real endpoint moving the right way, which is why the class is reported rather than dismissed. Three findings from the full text stop it short of open, and they were checked precisely because vitamin C is sold everywhere.

1. **BUBR1 was never measured after the agent.** The paper's own numbered conclusions list the BUBR1 deficit as part of the knockout characterisation and the ascorbate effect only as a reduction in segregation errors. The discussion attributes that effect to chemistry rather than to BUBR1: "AsA can directly scavenge superoxide molecules and reduce H2O2 to H2O via the ascorbate peroxidase reaction". The proposed mechanism, restoring BUBR1, is not what was measured.
2. **NAC did the same thing at the same concentration**, and the authors say they expected it not to, because "the rate constants for the reactions of NAC with superoxide are relatively low". The active principle reads as antioxidant capacity in general rather than ascorbate specifically, which is a different and weaker claim than a BUBR1-directed one.

3. **The model is an induced-oxidative-stress knockout**, so the stress-dependence clause of the frozen exposure rule applies, exactly as it does to the mouse model behind the chaperone co-inducer in class 14. An agent that corrects a lesion it was given does not thereby correct a different lesion.

Failed search route for the missing datum: a query pairing BubR1 or *BUB1B* with ascorbic acid, vitamin C, antioxidant, oxidative stress or N-acetylcysteine returns 22 records, titles read, and **none measures BUBR1 protein or kinetochore occupancy after an antioxidant**. **Deciding experiment:** repeat the 500 uM ascorbate arm and measure BUBR1 kinetochore recruitment and total protein after the agent in the same oocytes, then repeat in a *BUB1B*-low model that carries no oxidative-stress lesion. The second half is what separates "antioxidants fix oxidative damage" from "antioxidants raise BUBR1".

Appendix C. Register of tests run

Every test, screen, check and attempt behind this report, including the ones that returned nothing, were never run, or could not be answered. It exists because a test left out of a report reads as a test that was never done. The twenty-one pre-registered entries with their rules, hashes and results are in Section 12 and are not repeated here; this table adds the work that Section 12 does not carry, and points at the section for the work it does.

Where confirming an item needs something outside the repository, the line says what it needs.

#	Test	What was run	Verdict	Where, and what confirming it needs
1	The twenty-one pre-registered ledger entries	Each with its decision rule fixed and hashed before its inputs were opened	All run as recorded, with four instrument failures among them reported as failures	Section 12, with the full table of rules, results and file hashes. Needs the public data sources named in Section 12.2
2	Mechanism-class screen, class level	A two-arm pre-registered universe over a fixed 46-symbol gene set, grouped into twenty-four classes giving twenty-five verdict rows, with sixty DepMap targets measured	11 excluded, 6 not tested, 8 open. Two of our own arms moved down. Five limits disclosed	Sections 15 and Appendix B. Needs DGIdb, ChEMBL, PubMed and the public DepMap files
3	Mechanism-class screen, drug level	**181 molecules evaluated one by one against the frozen criteria, in five batches of 20, 14, 49, 49 and 49**	**Zero kept.** 29 removed on a named criterion, 151 UNRESOLVED because no source demonstrates the proposed mechanism on BUBR1 at any concentration, and the one provisionally kept was later closed as MECHANISM NOT TESTED	Section 15.1. Needs the same sources as item 2, plus openFDA and the EMA public pages
4	Mechanism-class screen, proteostasis slice	**Never run.** The remaining 45 molecules, covering HSP90 inhibition, chaperone co-induction and chemical chaperones, were never evaluated drug by drug	COVERAGE GAP, and it falls on two of the three arms carried forward in Section 7	Sections 7.2, 7.3, 13 and 15.6. Closing it needs the same sources as item 3 and a run that was not performed

#	Test	What was run	Verdict	Where, and what confirming it needs
5	Tissue co-expression of *BUB1B* and the NAD effector genes (`T2-GTEX-01`)	The query was recorded before the run and then issued against the public expression API under two dataset versions, with versioned and unversioned identifiers	**COULD NOT BE ANSWERED.** Every call returned zero rows; the endpoint or the dataset parameter was never resolved, and the arm is not counted as a result. The tissue-expression question behind the SIRT2 route is therefore still open, and it overlaps the untested NAPRT question in Section 7.4	This appendix. Needs the expression API and a correct endpoint, which we did not establish
6	Outside critiques of the proposal	Two de-identified questions put to outside models, neither of which saw any project data	Both said the proposal was not fundable as framed and that the NAD+ claim had to be narrowed, which is criticism we went looking for and are reporting. **One model returned quantitative figures that are not in the source it attributed them to; those figures were discarded and appear nowhere in this report.** A gene-disease curation page could not be retrieved and is used nowhere	This appendix and Section 12.2. Needs nothing; the discard is recorded so it can be checked against the cited source
7	Citation work, four passes	A working ledger check, a record-level check of the cited identifiers, two independent audits of this report with the second re-deriving from primary sources before seeing the first, and an adversarial check of every identifier	One cited source was found not to support the sentence it was attached to and that claim was withdrawn; one source was found mischaracterised against our own interest and corrected; four errata on cited papers were found, three of them previously undisclosed	Sections 4.5, 8.3 and 12, and the References. Needs PubMed, PMC and Crossref
8	Adversarial simulated panel	Seven reviews over the previous revision, three seats run twice as blind replicates, plus a second-provider reviewer at maximum effort	Unanimous on four findings, all of which drove tests that were then pre-registered and run; two of those tests removed material the panel suggested adding, and three weakened our own prior results	Appendix A.3. Needs nothing
9	Variant-effect model calibration	Three protein models calibrated under a rule written before any model ran, on controls chosen without model scores	All three informative; the three disagree at the allele class in question. Reported in this report only as the earlier placement test that Section 3's gate supersedes	Section 3 and the Track 1 methods report, section 7. Needs the public weights, a GPU and ClinVar

#	Test	What was run	Verdict	Where, and what confirming it needs
10	Phase of the allele pair	Pre-registered read-backed phasing	**UNRESOLVED.** No sequencing template spans the gap between the two sites, which is why the trans configuration is stated here as inferred and not measured	Section 5. Needs an alignment this report does not use; resolving it needs long reads or parental genotypes
11	Function of the missense allele	Not testable without a wet assay	**NOT TESTABLE** by any route available here, which is the whole content of the gate in Section 3	Sections 3 and 10. Needs the laboratory experiment stated in Section 10
12	*TRIP13* as a statistical control	Used as a pre-named positive control in the DepMap correlation test, and separately established as carrying no coding call in the Track 1 work	Reaches the test's threshold as a control. It is itself an MVA gene, which is stated rather than left implicit	Sections 2 and 9.4. Needs the public DepMap files
13	Readout detection limit	Measured in the Track 1 work on an alignment, to establish what clonal fraction bulk short-read data of this kind can resolve	Measured, with a recorded consequence for Track 2: an aggregate whole-genome aneuploidy-rate endpoint of the kind this report would want is below that detection limit and would need a different method. **The figure is derived from the child's data, so it is reported in the Track 1 methods report and not here**, and what Section 2 says instead is that no mechanism here has a validated biomarker	Section 2, and the Track 1 methods report, section 4. Needs an alignment this report does not use
14	External-service record	A log of every call that left the local machines	Recorded. Only gene, protein, drug and mechanism terms, identifiers and public accessions were sent	Section 12.2. Needs nothing
15	Compute and cost audit	A read-only audit of the whole project, every figure marked measured or estimated	Reported, including the gap: **the Track 2 share was never apportioned and is stated as missing rather than estimated**	Section 17, Q10. Needs the tool logs

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Bibliographic records were retrieved by PMID from PubMed (NCBI E-utilities) and by DOI from Crossref on 2026-10-21. For the 4 entries cited in the text as author and year, the first author and year of the record were checked against the citation: 4 of 4 match.

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